

Course: Networked Control

Lectures III.2/3: Sampling, delays, and packet dropouts

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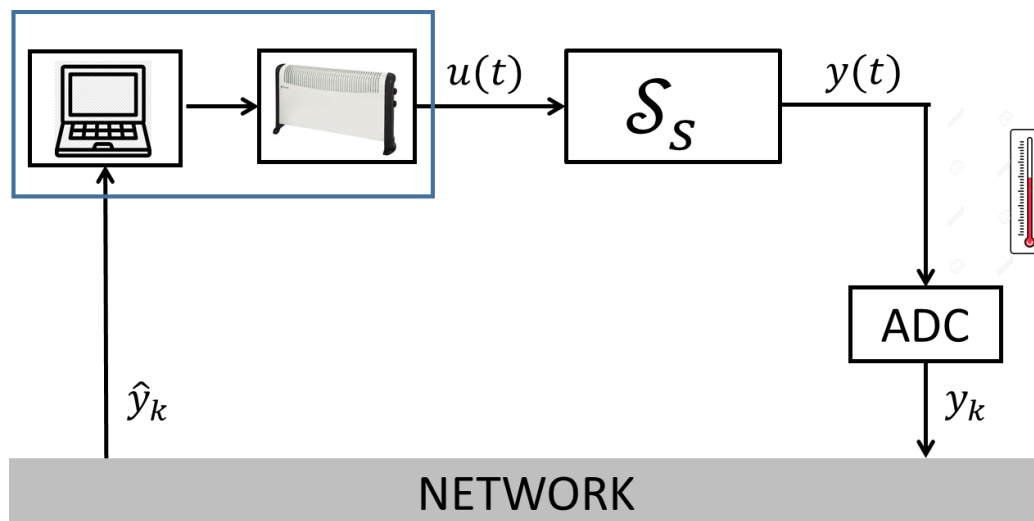
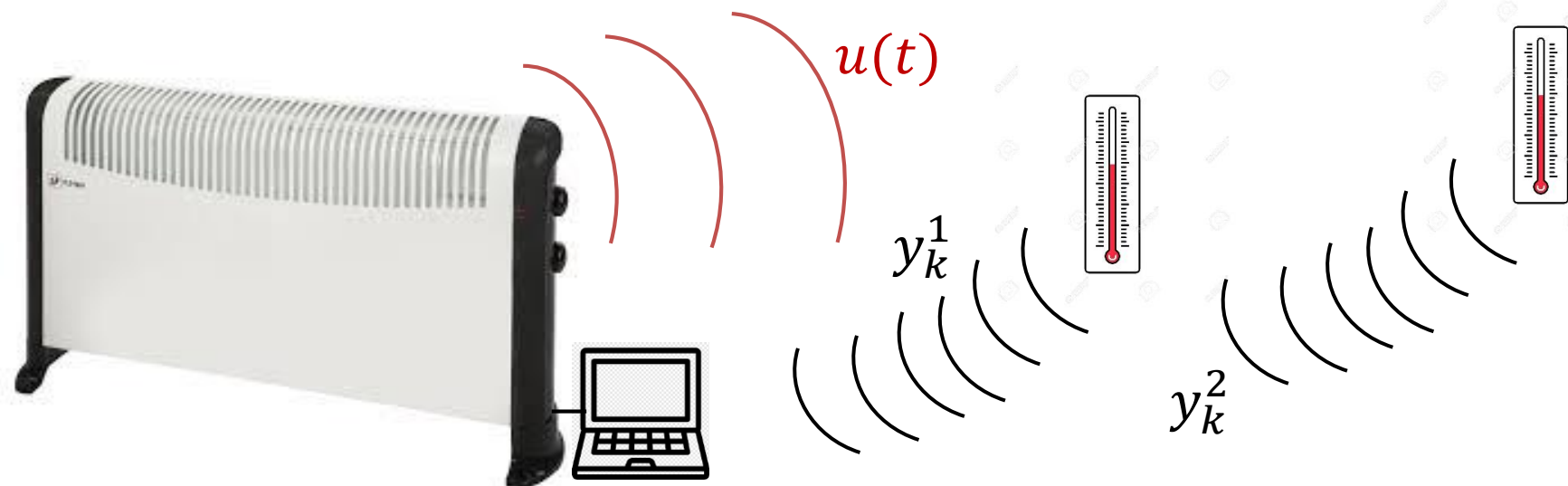
Outline

- Reference configuration
- Examples (constant delay)
- Examples (time-varying sampling time and delay)
- Compensation of delays
- Examples (constant sampling time and delay, with packet dropouts)

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- **Reference configuration**
- Examples (constant delay)
- Examples (time-varying sampling time and delay)
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- Examples (constant sampling time and delay, with packet dropouts)

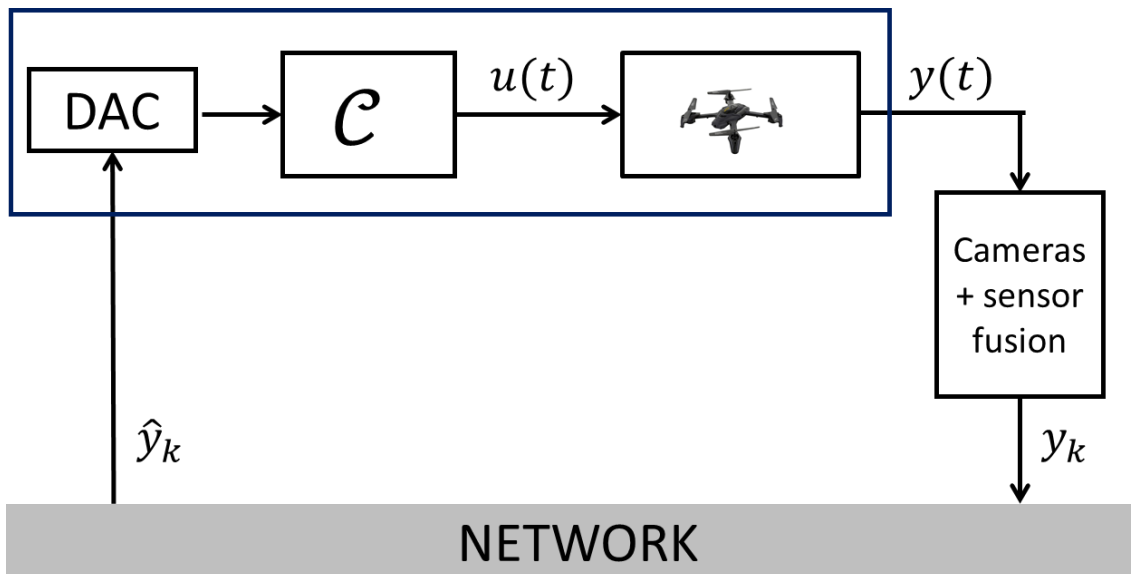
Reference configuration - examples



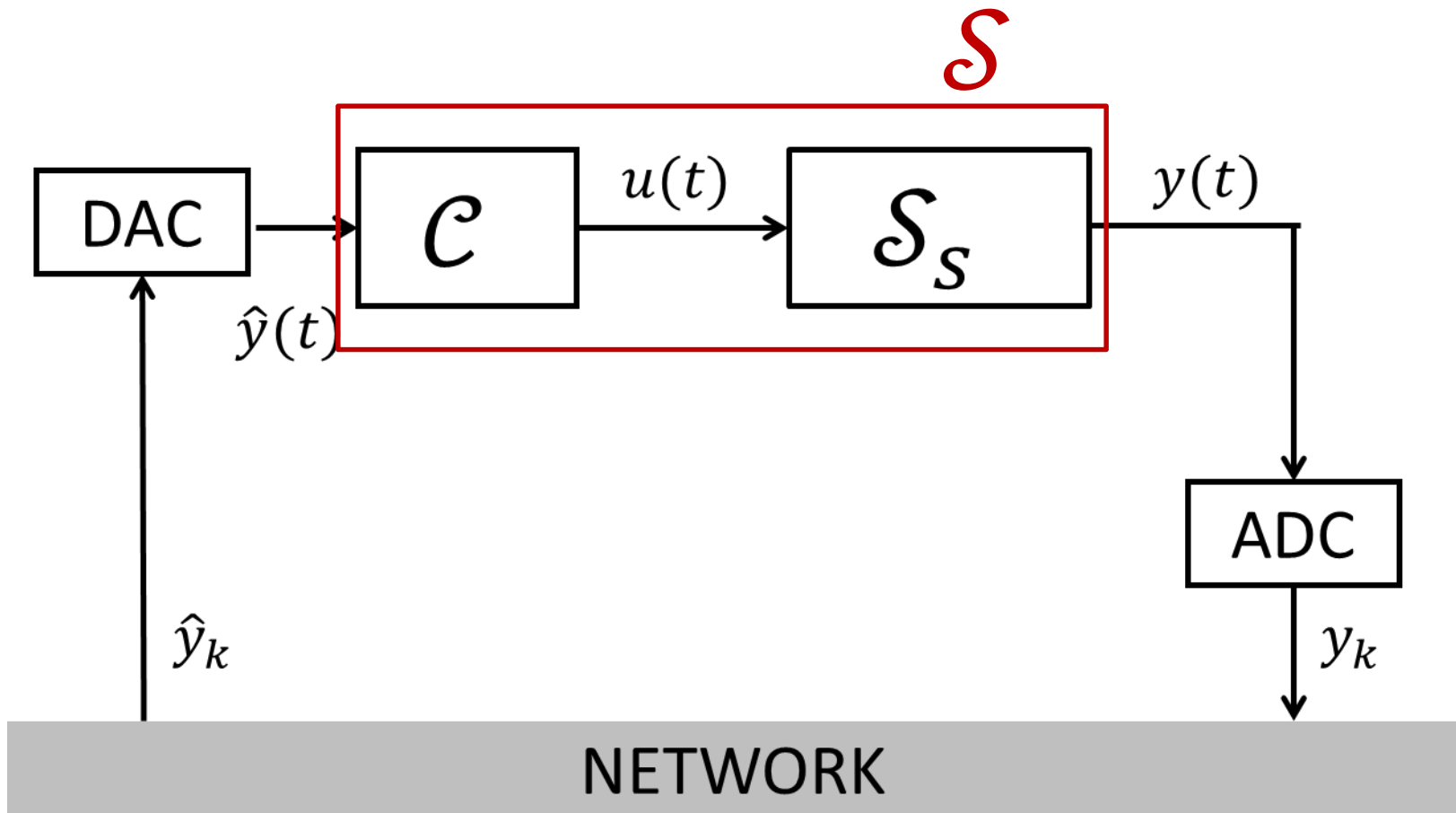
Reference configuration - examples

Flying arena

- Cameras (e.g., IR) sense distances
- Local transmission for sensor fusion (wired)
- Position y_k sent to the quadrotor on-board control station through wireless network

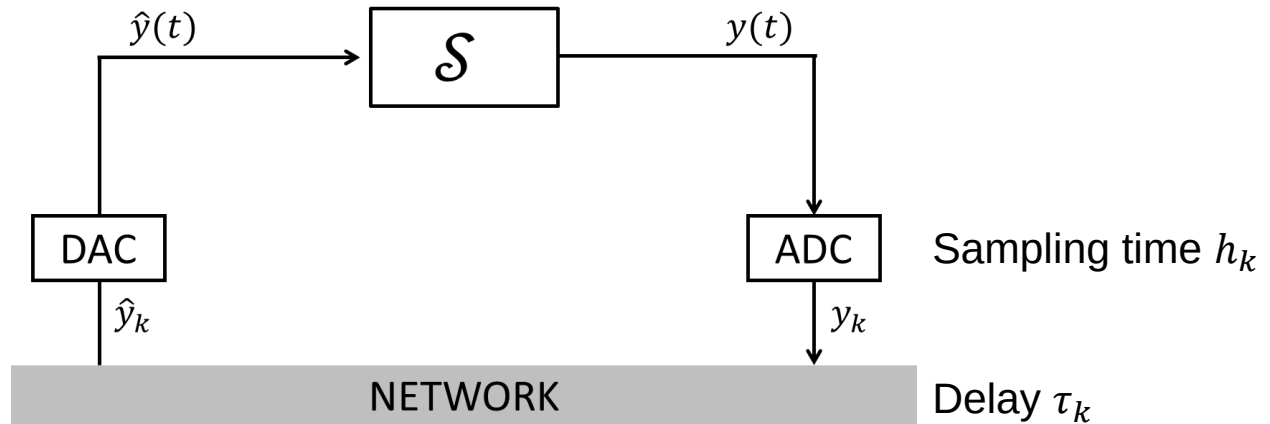


Reference configuration



Reference configuration

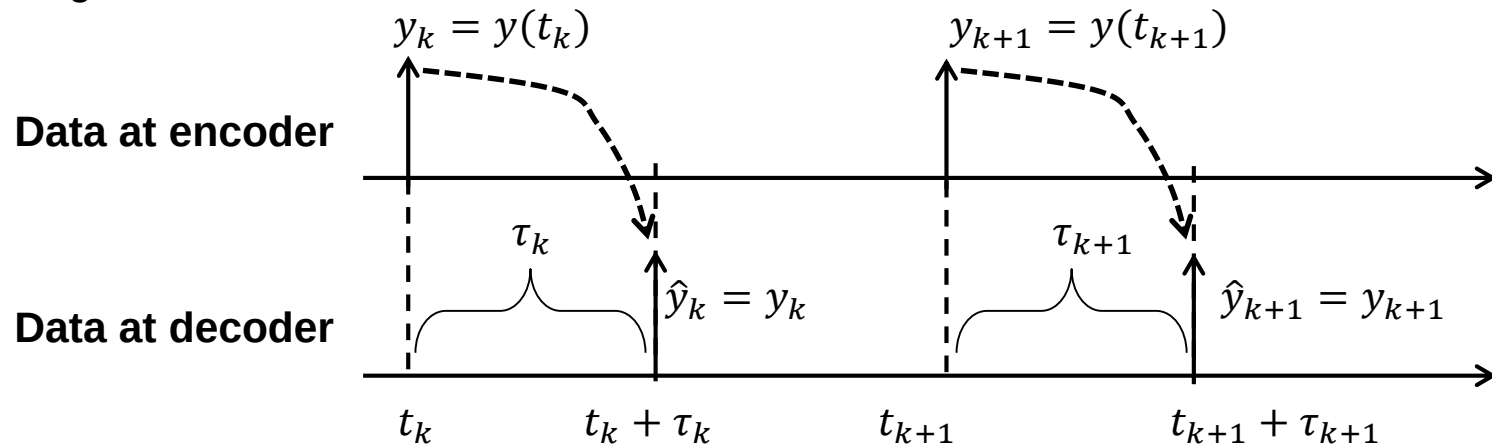
In this lecture we consider the general NCS scheme



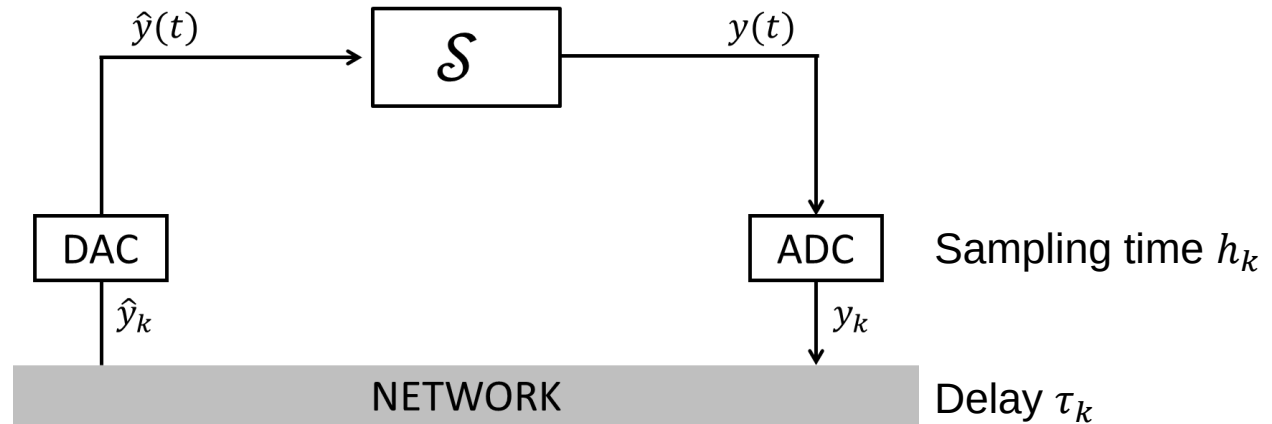
where the dynamics of $\mathcal{S}$ is

$$\begin{cases} \dot{x}(t) = Ax(t) + B\hat{y}(t) \\ y(t) = Cx(t) \end{cases}$$

Timing diagram



Reference configuration

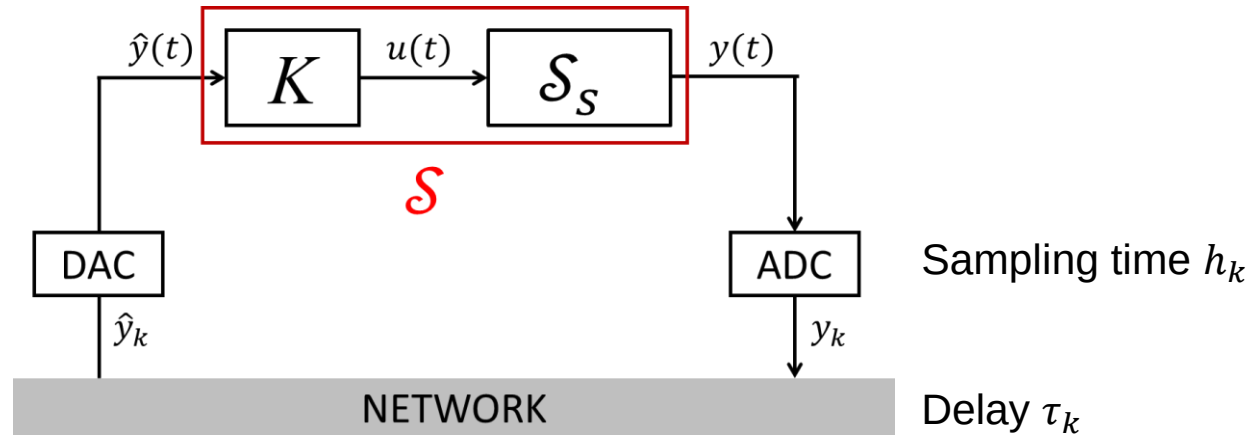


This scheme is very general/generalizable!

In fact we can describe, through the above scheme (and by proper reformulations), a number of different notable configurations (control systems), e.g.,

- Static and dynamic output-feedback controller co-located with the actuator
- Static and dynamic output-feedback controller co-located with the sensor
- Remote state-feedback controller
- ...

The case of static output-feedback control co-located with the actuator



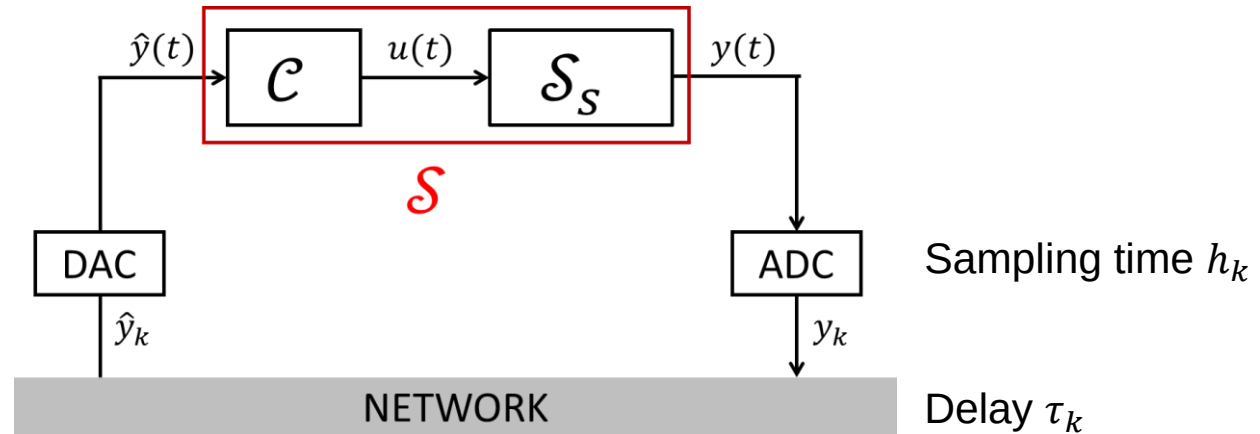
The dynamics of $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) &= A_S x_S(t) + B_S u(t) \\ y(t) &= C_S x_S(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. Renaming $x(t) = x_S(t)$, $A = A_S$, $B = B_S K$, $C = C_S$, we retrieve the scheme of the general configuration, i.e.,

$$\begin{cases} \dot{x}(t) &= Ax(t) + B\hat{y}(t) \\ y(t) &= Cx(t) \end{cases}$$

The case of dynamic output-feedback control co-located with the actuator



The dynamics of $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) &= A_S x_S(t) + B_S u(t) \\ y(t) &= C_S x_S(t) \end{cases}$$

and the dynamics of the dynamic output-feedback controller $\mathcal{C}$ is

$$\begin{cases} \dot{x}_C(t) &= A_C x_C(t) + B_C \hat{y}(t) \\ u(t) &= C_C x_C(t) + D_C \hat{y}(t) \end{cases}$$

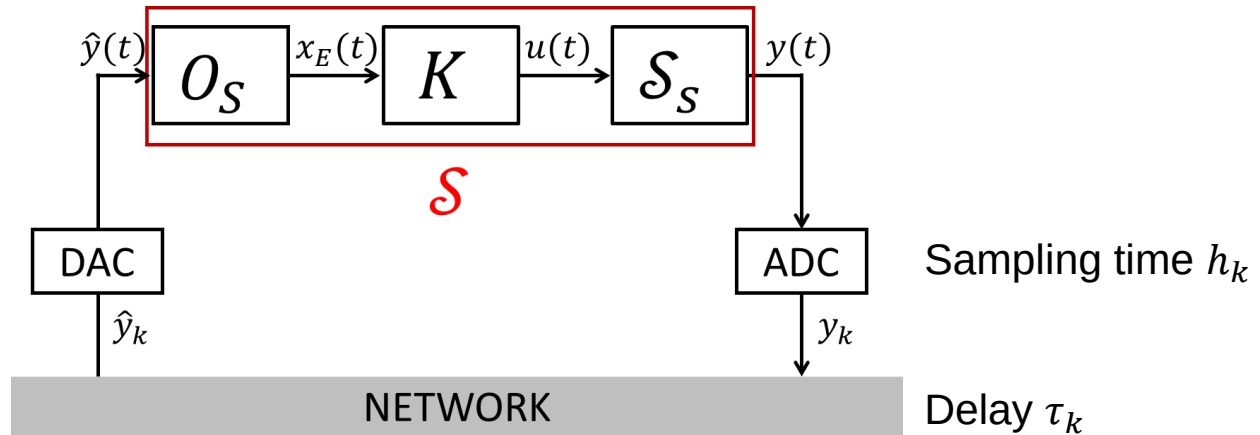
Overall, we write

$$\begin{cases} \dot{x}_S(t) &= A_S x_S(t) + B_S (C_C x_C(t) + D_C \hat{y}(t)) \\ \dot{x}_C(t) &= A_C x_C(t) + B_C \hat{y}(t) \end{cases}$$

Renaming $x(t) = \begin{bmatrix} x_S(t) \\ x_C(t) \end{bmatrix}^T$, $A = \begin{bmatrix} A_S & B_S C_C \\ 0 & A_C \end{bmatrix}$, $B = \begin{bmatrix} B_S D_C \\ B_C \end{bmatrix}$, $C = [C_S \quad 0]$, we obtain

$$\begin{cases} \dot{x}(t) &= Ax(t) + B \hat{y}(t) \\ y(t) &= Cx(t) \end{cases}$$

The case of observer + static state-feedback control co-located with the actuator



The dynamics of S_S is

$$\begin{cases} \dot{x}_S(t) &= A_S x_S(t) + B_S u(t) \\ y(t) &= C_S x_S(t) \end{cases}$$

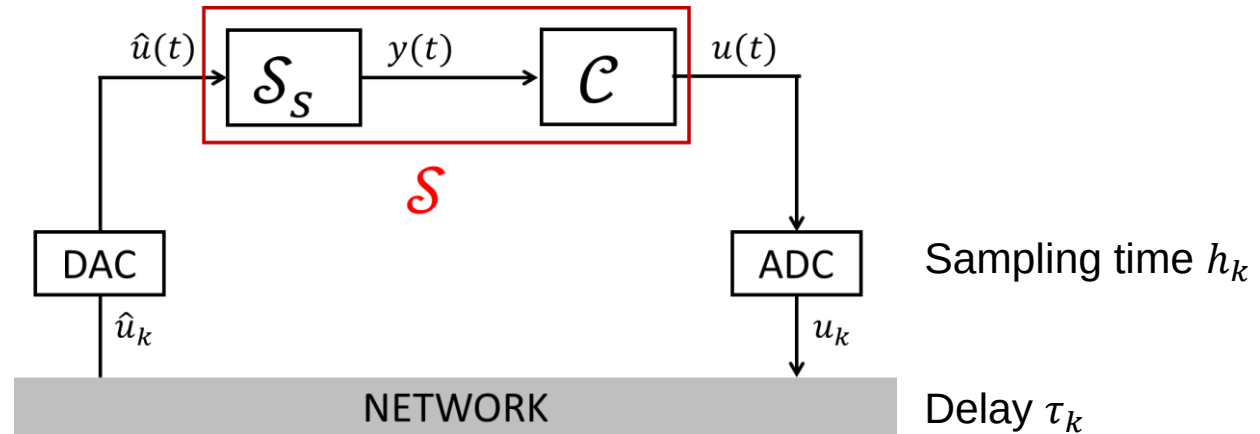
the dynamics of the observer O_S is, e.g., $\dot{x}_E(t) = A_S x_E(t) + B_S u(t) + L_E(\hat{y}(t) - C_S x_E(t))$, and the control action is set to $u(t) = K x_E(t)$. Overall, we write

$$\begin{cases} \dot{x}_S(t) &= A_S x_S(t) + B_S K x_E(t) \\ \dot{x}_E(t) &= A_S x_E(t) + B_S K x_E(t) + L_E(\hat{y}(t) - C_S x_E(t)) \end{cases}$$

Renaming $x(t) = \begin{bmatrix} x_S(t) \\ x_E(t) \end{bmatrix}^T$, $A = \begin{bmatrix} A_S & B_S K \\ 0 & A_S + B_S K - L_E C_S \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ L_E \end{bmatrix}$, $C = [C_S \quad 0]$, we obtain

$$\begin{cases} \dot{x}(t) &= A x(t) + B \hat{y}(t) \\ y(t) &= C x(t) \end{cases}$$

The case of output-feedback control co-located with the sensor



The dynamics of $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) &= A_S x_S(t) + B_S \hat{u}(t) \\ y(t) &= C_S x_S(t) \end{cases}$$

and the dynamics of the dynamic output-feedback controller $\mathcal{C}$ is

$$\begin{cases} \dot{x}_C(t) &= A_C x_C(t) + B_C y(t) \\ u(t) &= C_C x_C(t) + D_C y(t) \end{cases}$$

Overall, we write

$$\begin{cases} \dot{x}_S(t) &= A_S x_S(t) + B_S \hat{u}(t) \\ \dot{x}_C(t) &= A_C x_C(t) + B_C C_S x_S(t) \end{cases}$$

Renaming $x(t) = \begin{bmatrix} x_S(t) \\ x_C(t) \end{bmatrix}^T$, $A = \begin{bmatrix} A_S & 0 \\ B_C C_S & A_C \end{bmatrix}$, $B = \begin{bmatrix} B_S \\ 0 \end{bmatrix}$, $C = [D_C C_S \quad C_C]$, we obtain

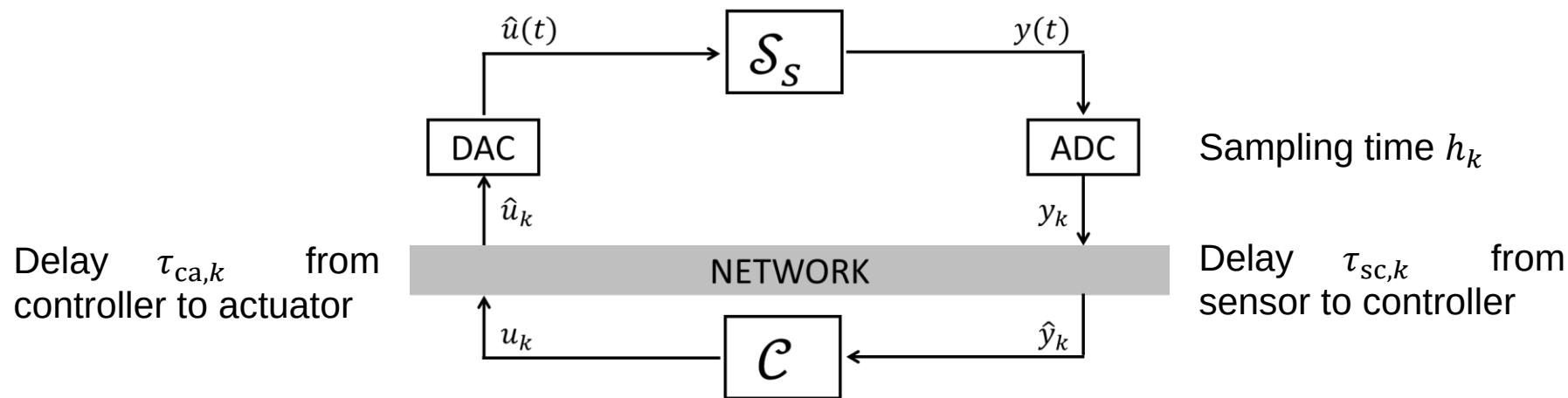
$$\begin{cases} \dot{x}(t) &= Ax(t) + B\hat{u}(t) \\ u(t) &= Cx(t) \end{cases} \Rightarrow$$

Original scheme where $y(t) \rightarrow u(t)$ and $\hat{y}(t) \rightarrow \hat{u}(t)$

Remarks:

- Setting $C_C = 0$, $D_C = K$ this is the static output-feedback scheme
- With suitable definition of C_C and D_C also the scheme with the observer is considered

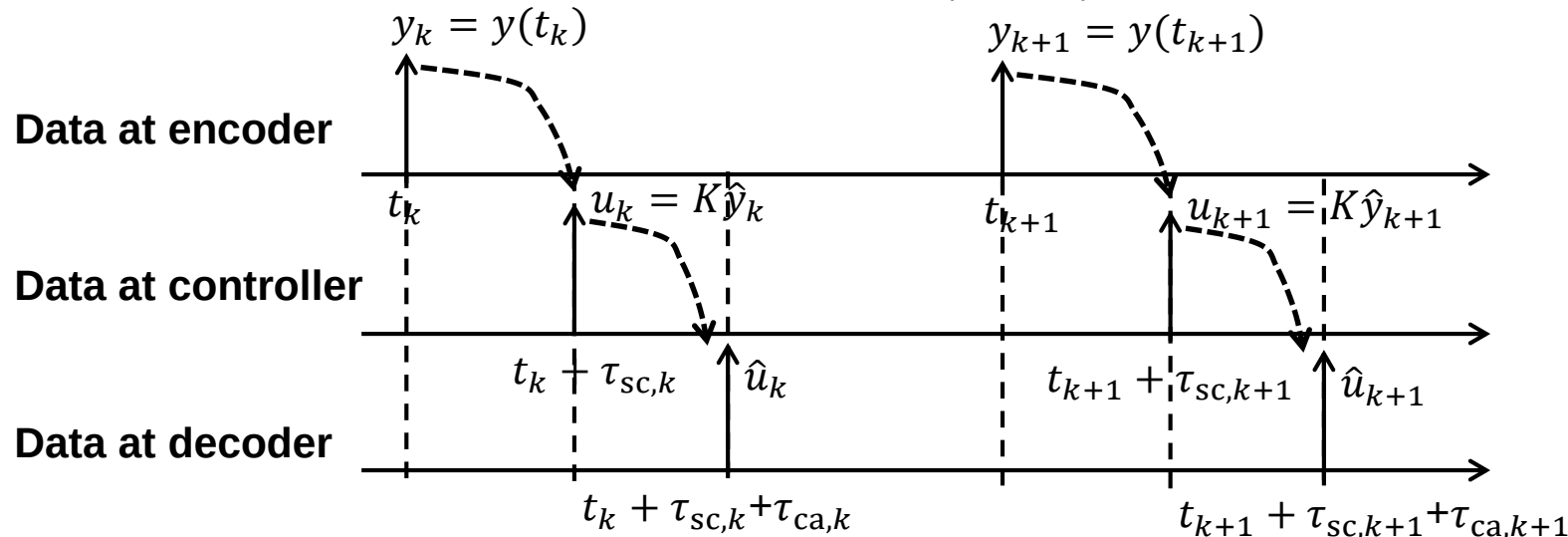
The case of **output-feedback** **remote** control



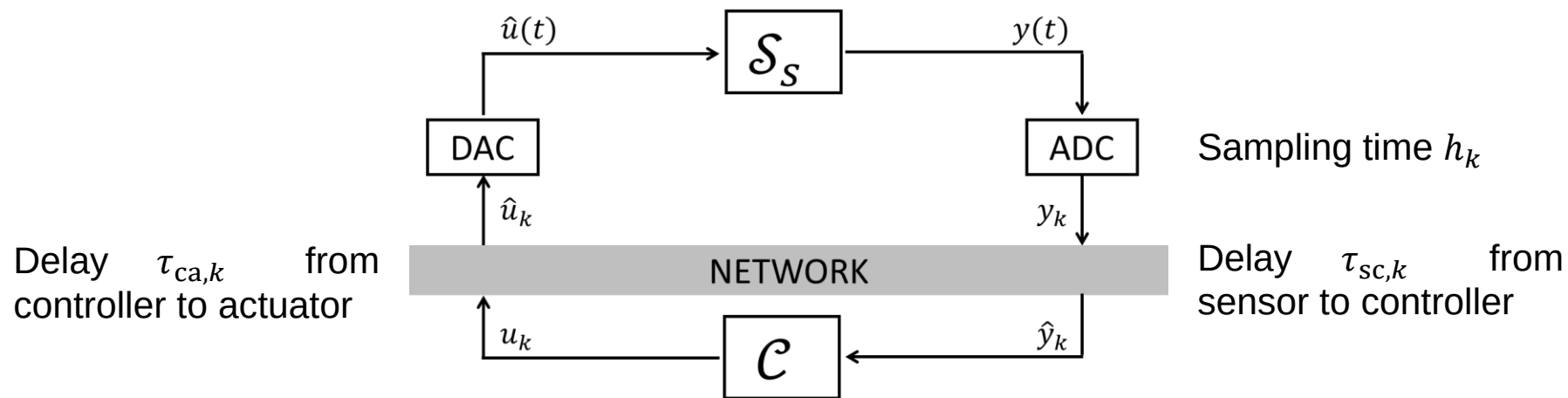
The dynamics of $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) = A_S x_S(t) + B_S u(t) \\ y(t) = C_S x_S(t) \end{cases}$$

An output-feedback static controller is used, i.e., $u_k = K \hat{y}_k$.



The case of **output-feedback static** remote control

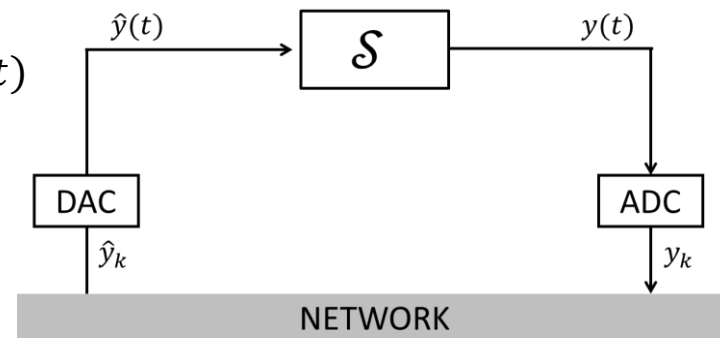


Renaming $x(t) = x_S(t)$, $A = A_S$, $B = B_S K$, $C = C_S$, this scheme is equivalent to the one of the general configuration, i.e.,

$$\begin{cases} \dot{x}(t) = Ax(t) + B\hat{y}(t) \\ y(t) = Cx(t) \end{cases}$$

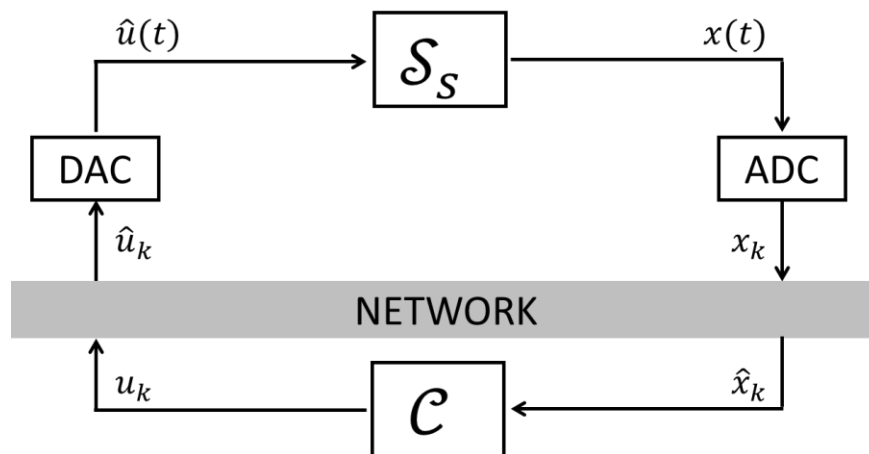
provided that the (unique) delay is set to

$$\tau_k = \tau_{sc,k} + \tau_{ca,k}$$

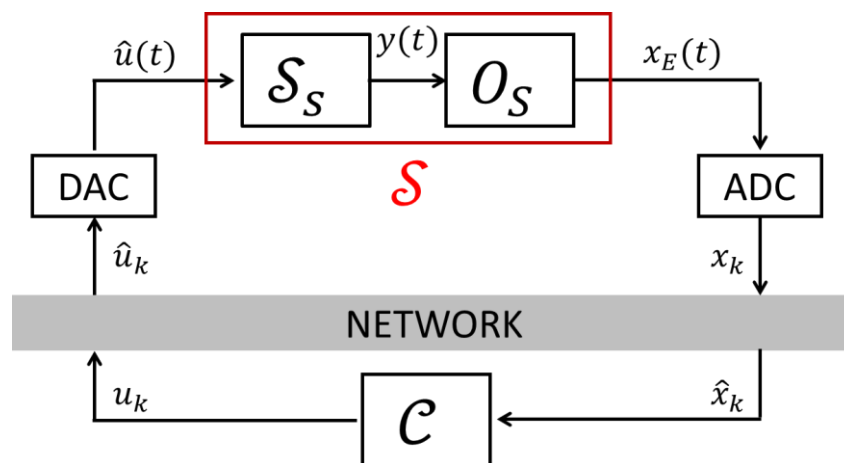


The case of **general-type** remote control

If $C_S = I$ a static state-feedback controller is obtained

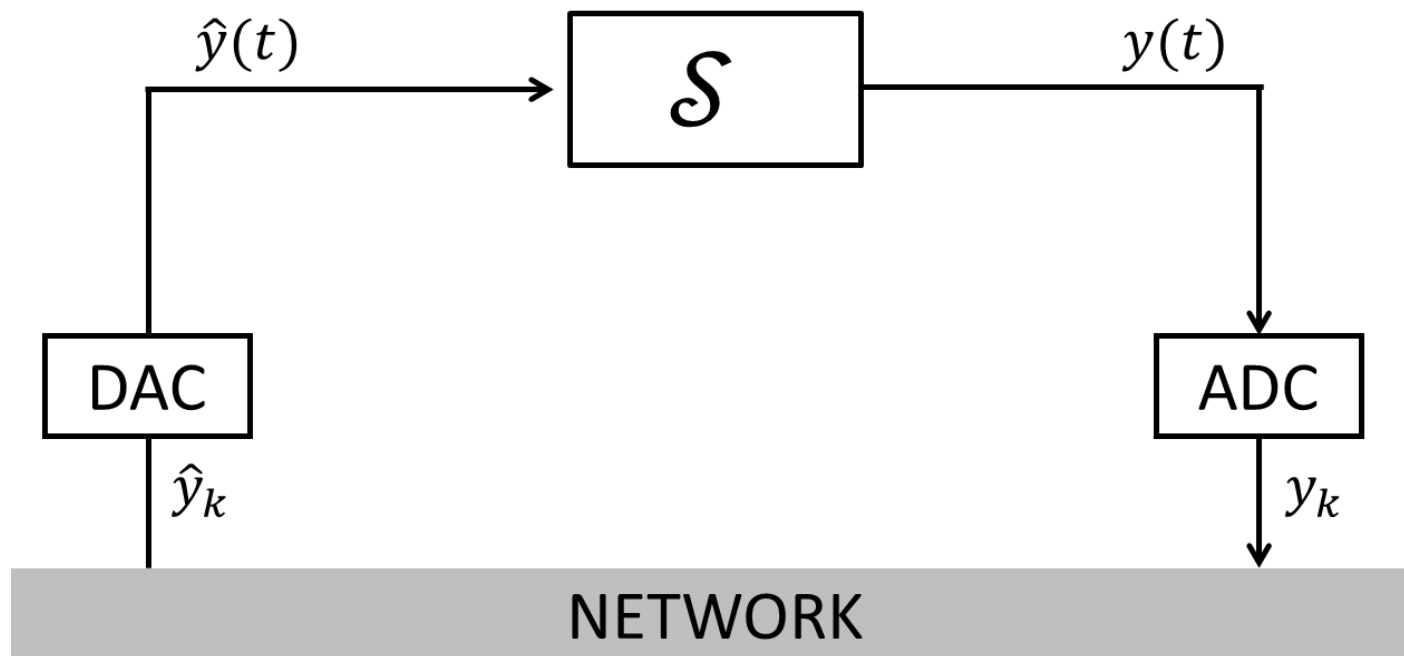


If $\mathcal{S}$ includes both system and state observer, also the case of dynamic output-feedback control can be considered



Reference configuration

In conclusion, the scheme



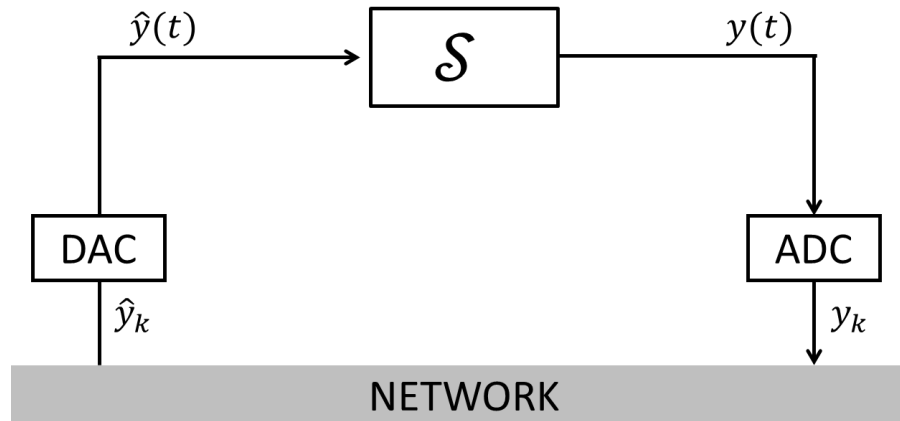
is a suitable abstraction for an extremely wide variety of control systems, among which the ones with a remote controller

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- **Examples (constant delay)**
- Examples (time-varying sampling time and delay)
- Compensation of delays
- Examples (constant sampling time and delay, with packet dropouts)

Examples: summary

We consider the general NCS scheme



where the dynamics of $\mathcal{S}$ is

$$\begin{cases} \dot{x}(t) &= Ax(t) + B\hat{y}(t) \\ y(t) &= Cx(t) \end{cases}$$

Under Assumption 1 (constant sampling time and delay), the dynamics of the NCS is described by

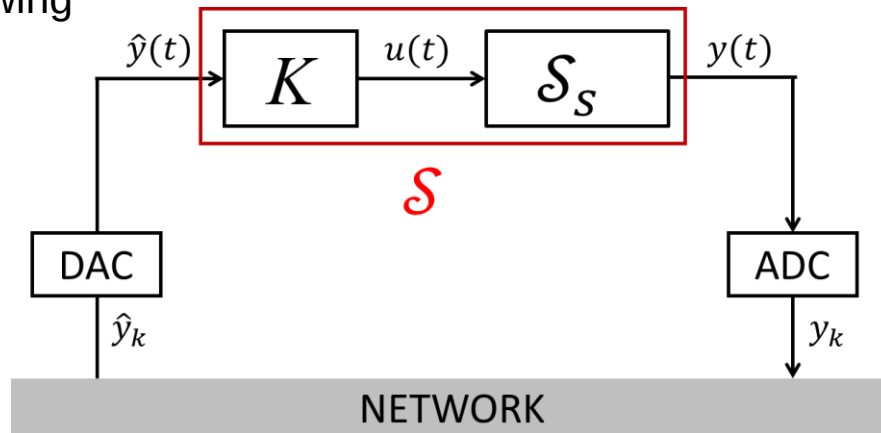
$$z_{k+1} = \Phi(h, \tau)z_k$$

where $z_k = [x_k, \hat{y}_{k-1}]^T$ and

$$\Phi(h, \tau) = \begin{bmatrix} e^{Ah} + \Gamma(h - \tau)BC & e^{A(h-\tau)}\Gamma(\tau)B \\ C & 0 \end{bmatrix}, \Gamma(\xi) = \int_0^\xi e^{Av} dv$$

Examples: summary

In case system $\mathcal{S}$ consists of system+output feedback controller co-located with the actuator, the scheme is the following



where the dynamics of $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) &= A_S x_S(t) + B_S u(t) \\ y(t) &= C_S x_S(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. Renaming $x(t) = x_S(t)$, $A = A_S$, $B = B_S K$, $C = C_S$, we retrieve the scheme of the previous slide. So, we rewrite

$$\Phi(h, \tau) = \begin{bmatrix} e^{A_S h} + \Gamma(h - \tau) B_S K C_S & e^{A_S(h-\tau)} \Gamma(\tau) B_S K \\ C_S & 0 \end{bmatrix}, \Gamma(\xi) = \int_0^\xi e^{A_S v} dv$$

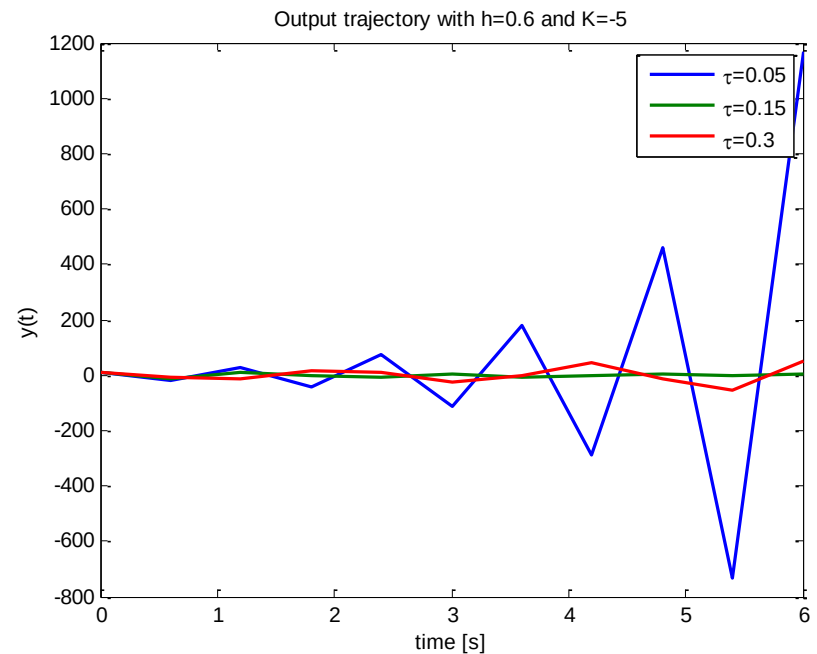
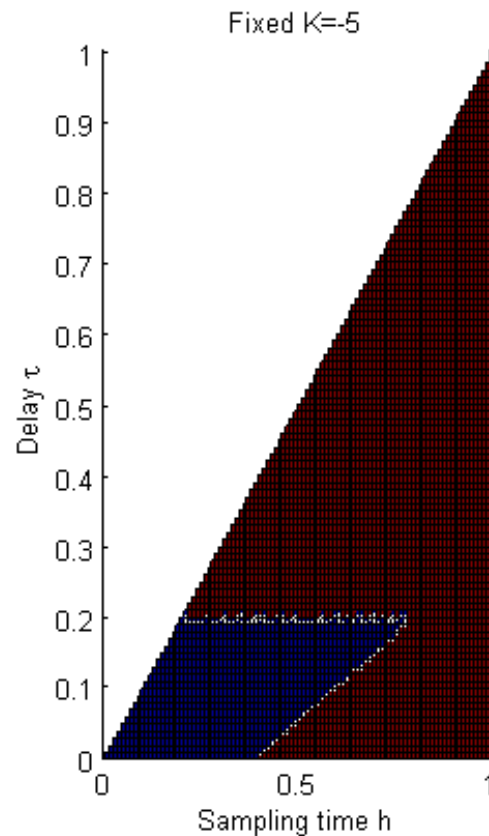
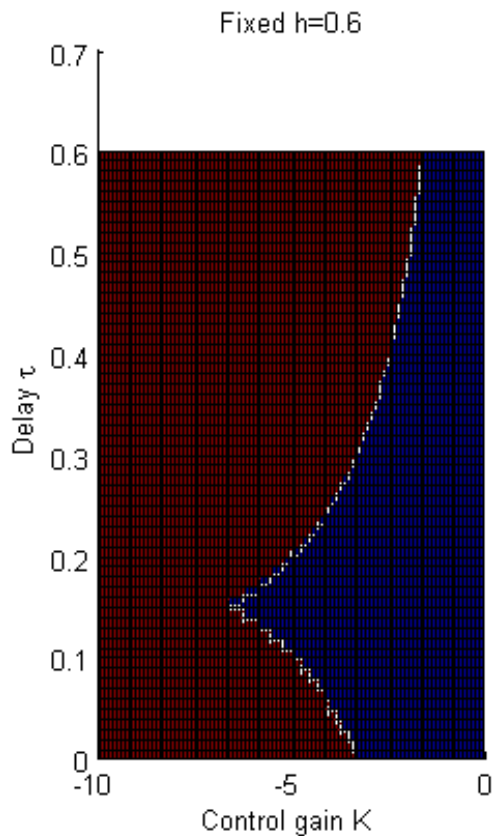
Example 1

The dynamics of the scalar system $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) &= u(t) \\ y(t) &= x_S(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\Phi(h, \tau) = \begin{bmatrix} 1 + K(h - \tau) & K\tau \\ 1 & 0 \end{bmatrix}$$



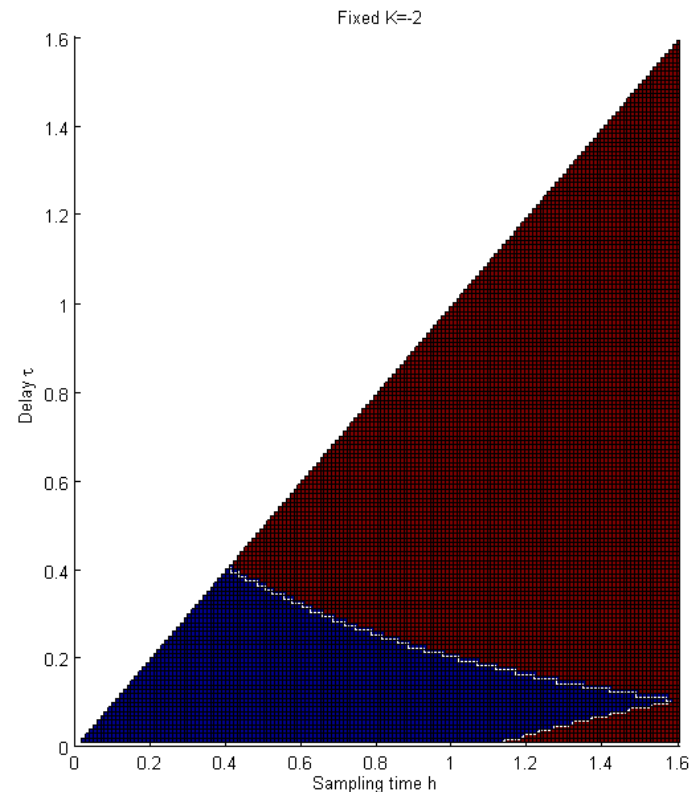
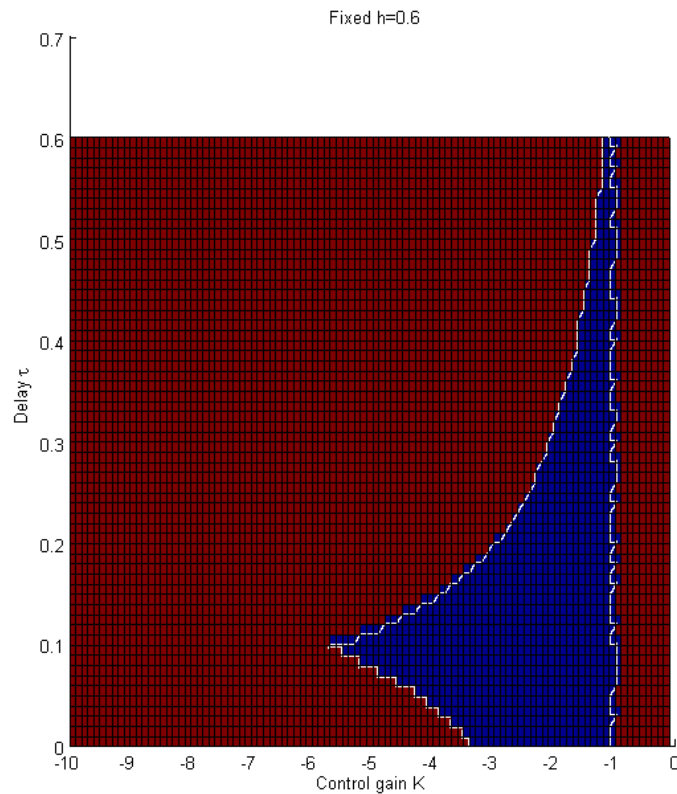
Example 2

The dynamics of the scalar system $\mathcal{S}_s$ is

$$\begin{cases} \dot{x}_s(t) &= x_s(t) + u(t) \\ y(t) &= x_s(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\Phi(h, \tau) = \begin{bmatrix} e^h + K(e^{h-\tau} - 1) & Ke^{h-\tau}(e^\tau - 1) \\ 1 & 0 \end{bmatrix}$$



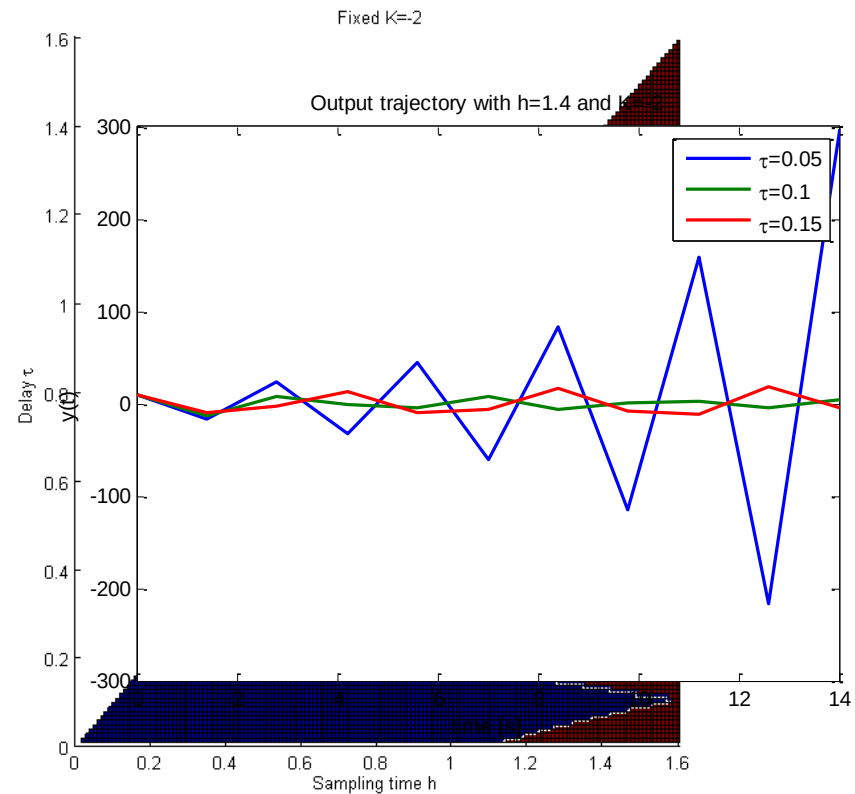
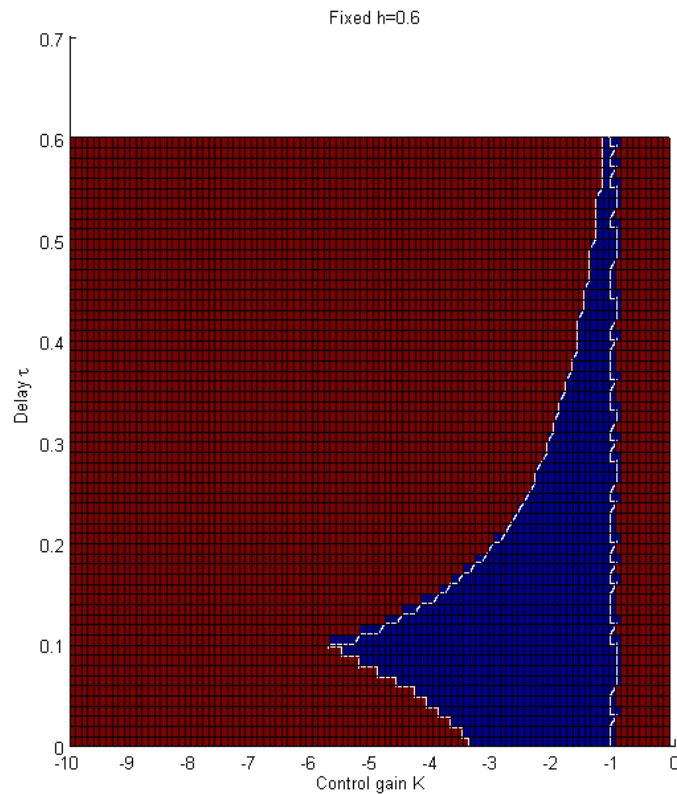
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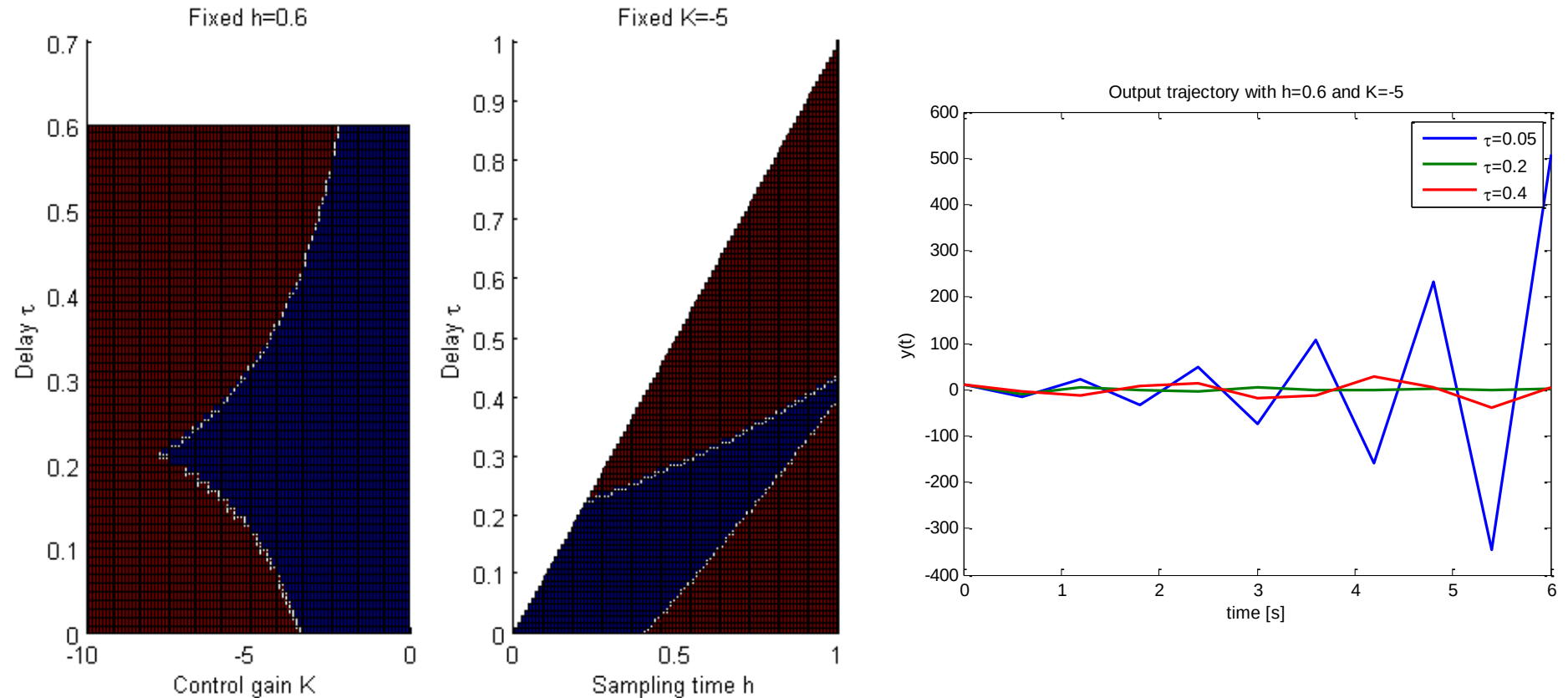
Example 3

The dynamics of the scalar system $\mathcal{S}_s$ is

$$\begin{cases} \dot{x}_s(t) &= -x_s(t) + u(t) \\ y(t) &= x_s(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\Phi(h, \tau) = \begin{bmatrix} e^{-h} + K(1 - e^{-(h-\tau)}) & Ke^{-(h-\tau)}(1 - e^{-\tau}) \\ 1 & 0 \end{bmatrix}$$



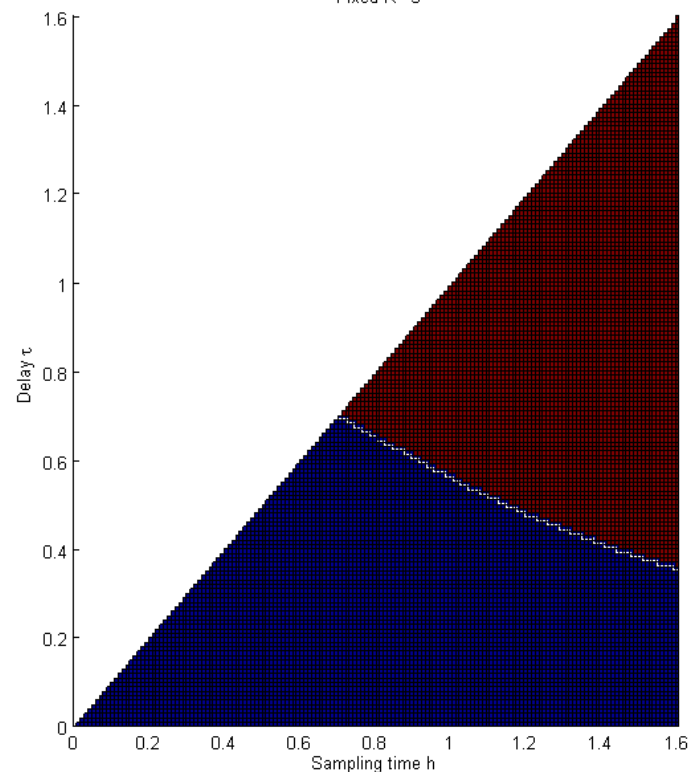
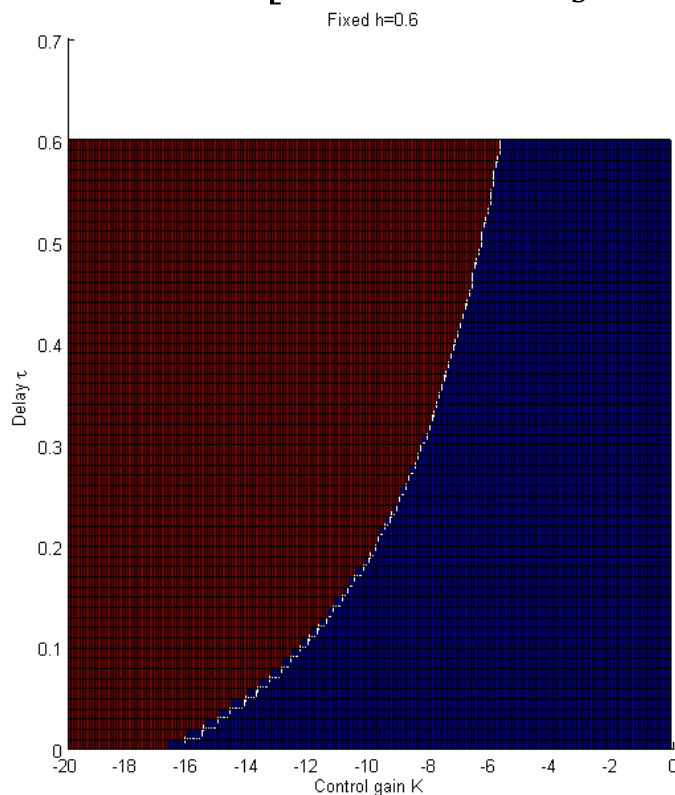
Example 4

The dynamics of the 2x2 system $\mathcal{S}$ is

$$\begin{cases} \dot{x}_S(t) = A_S x_S(t) + B_S u(t) \\ y(t) = C_S x_S(t) \end{cases}, A_S = \begin{bmatrix} -1 & 0.9 \\ 0.2 & -0.2 \end{bmatrix}, B_S = \begin{bmatrix} 1 \\ 0.1 \end{bmatrix}, C_S = [-0.1 \ 1]$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\Phi(h, \tau) = \begin{bmatrix} e^{A_S h} + A_S^{-1}(e^{A_S(h-\tau)} - I)B_S K C_S & e^{A_S(h-\tau)} A_S^{-1}(e^{A_S \tau} - I)B_S K \\ C_S & 0 \end{bmatrix}$$



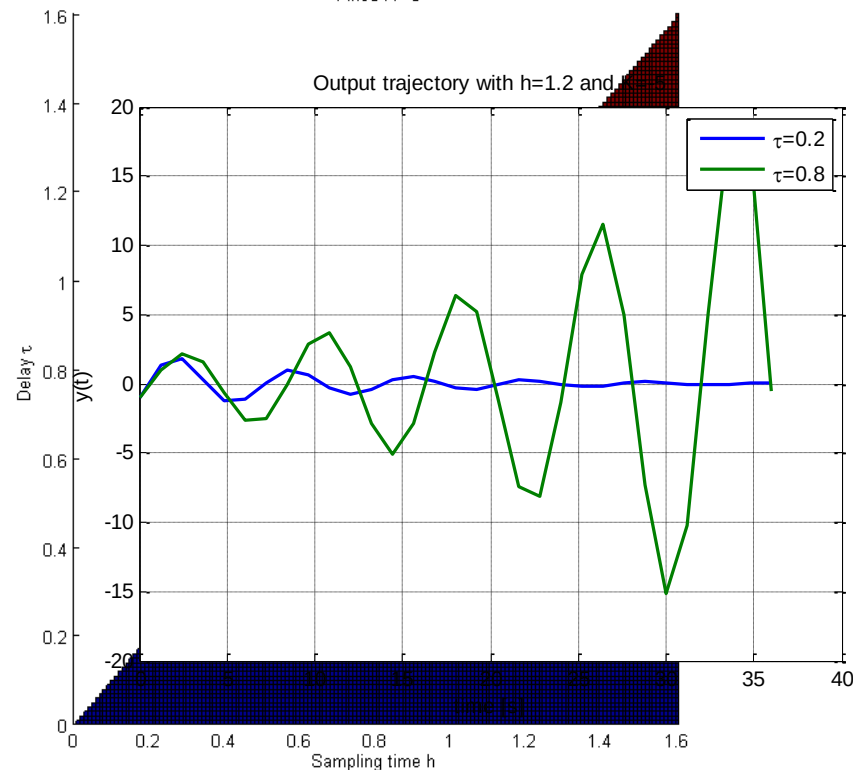
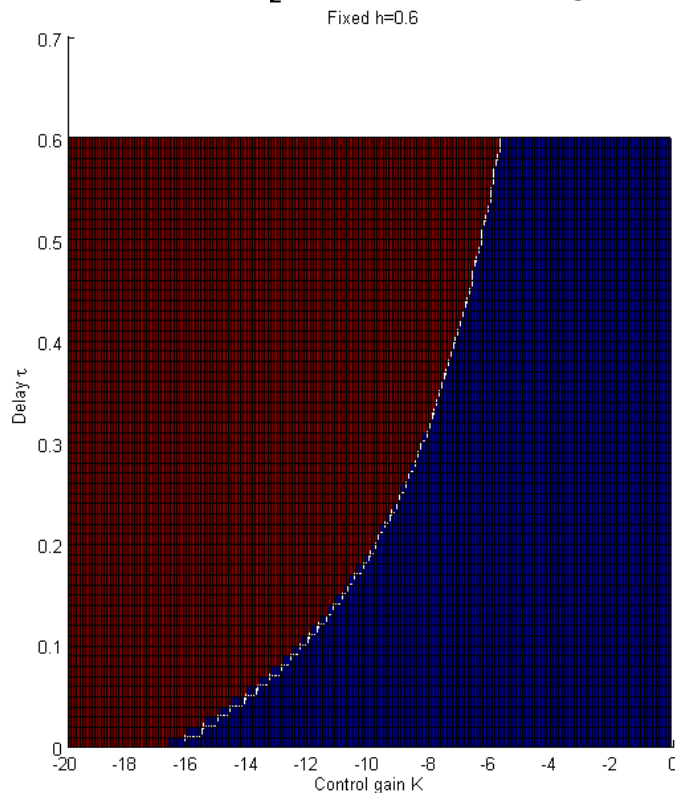
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and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

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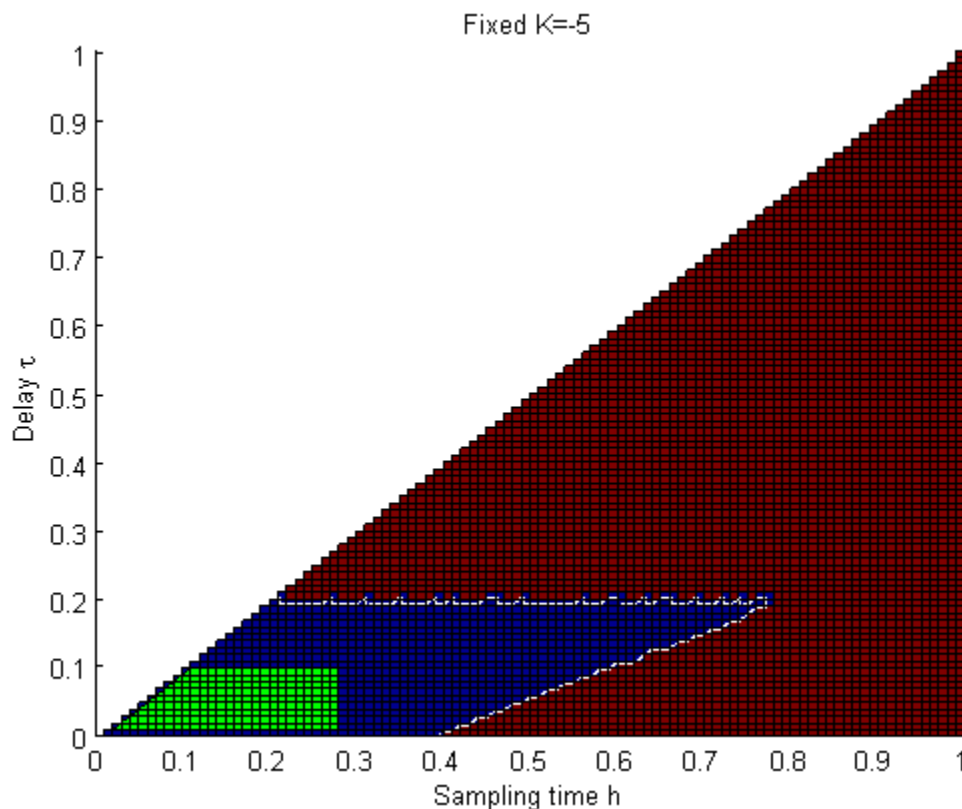
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$$\Phi(h_k, \tau_k) = \begin{bmatrix} 1 + K(h_k - \tau_k) & K\tau_k \\ 1 & 0 \end{bmatrix}$$



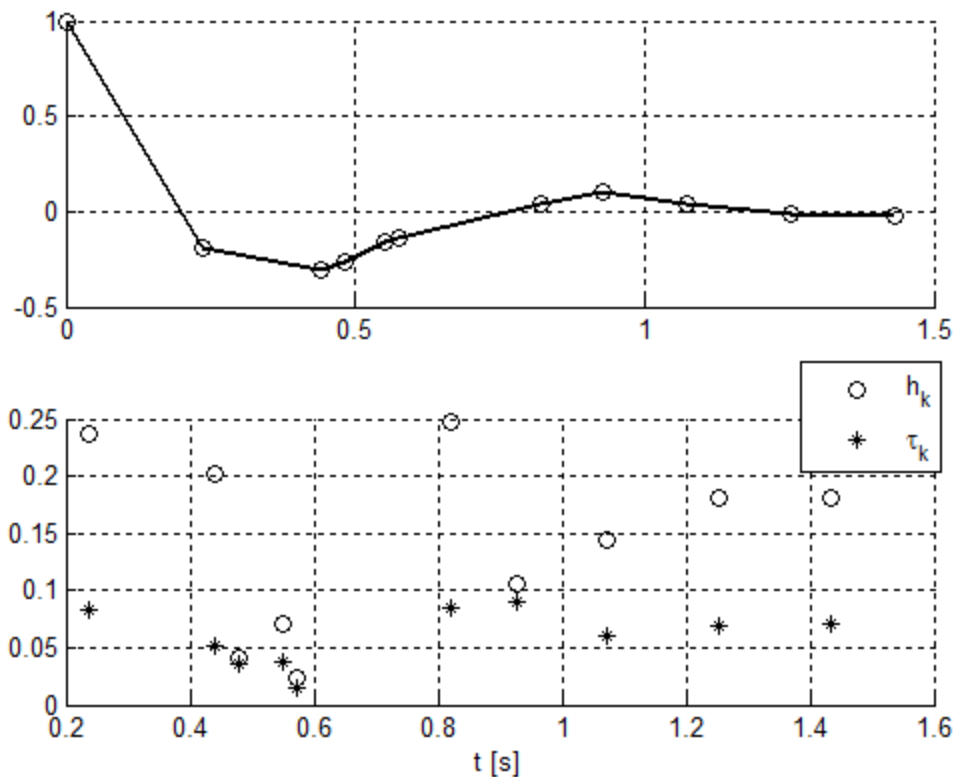
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Example 2

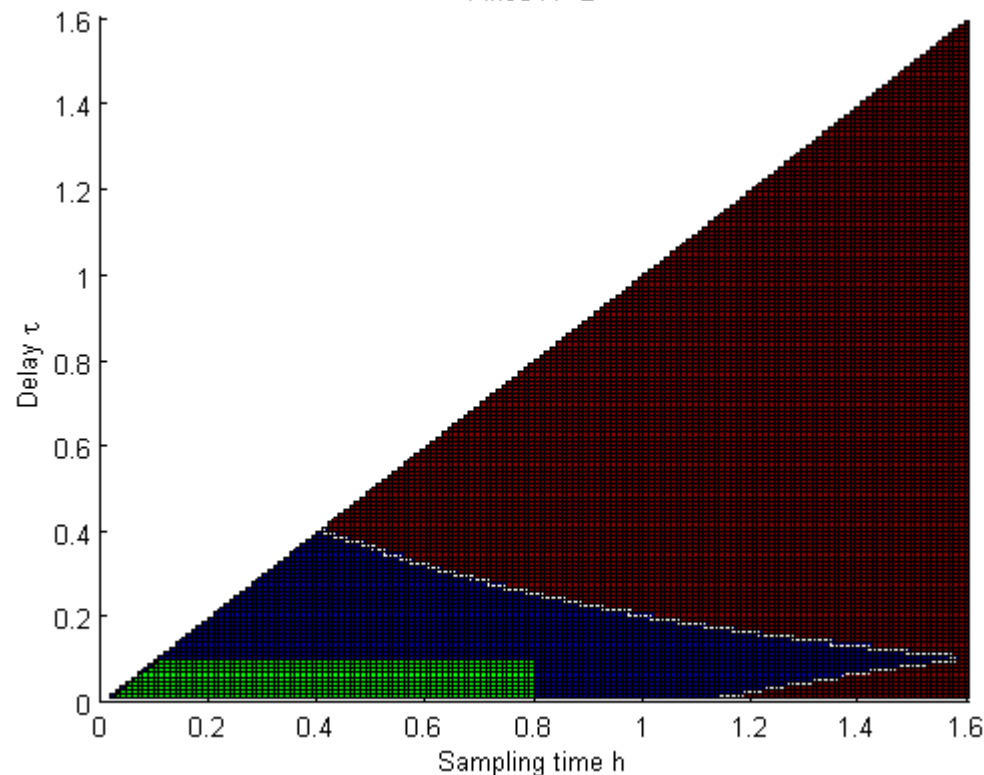
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and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\Phi(h_k, \tau_k) = \begin{bmatrix} e^{h_k} + K(e^{h_k - \tau_k} - 1) & Ke^{h_k - \tau_k}(e^{\tau_k} - 1) \\ 1 & 0 \end{bmatrix}$$

Fixed $K=-2$



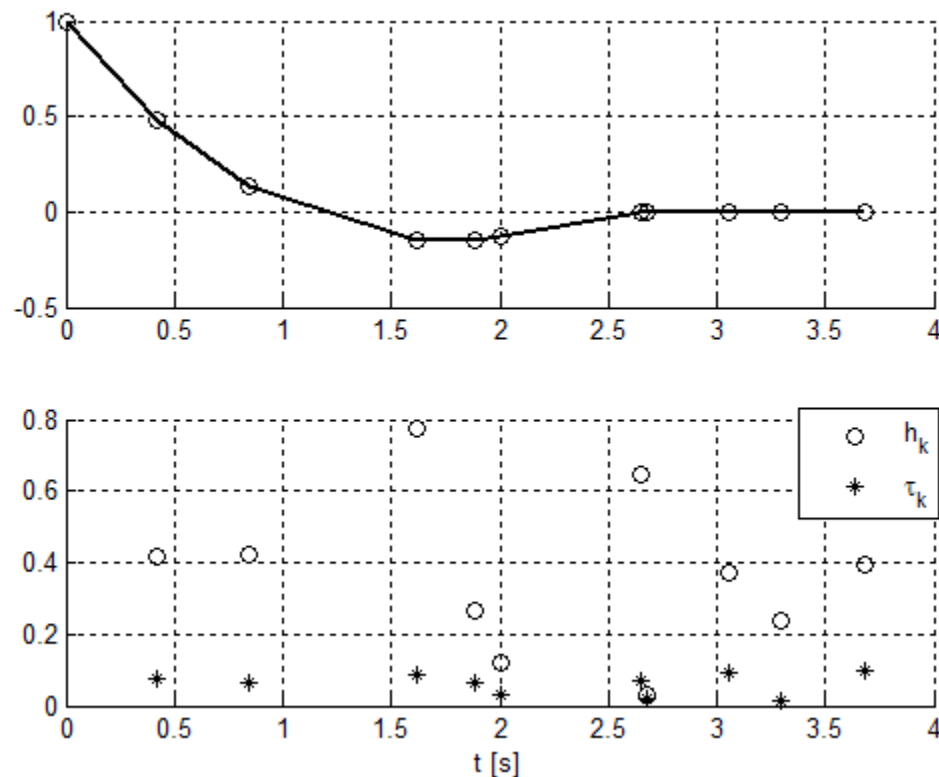
Example 2

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and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

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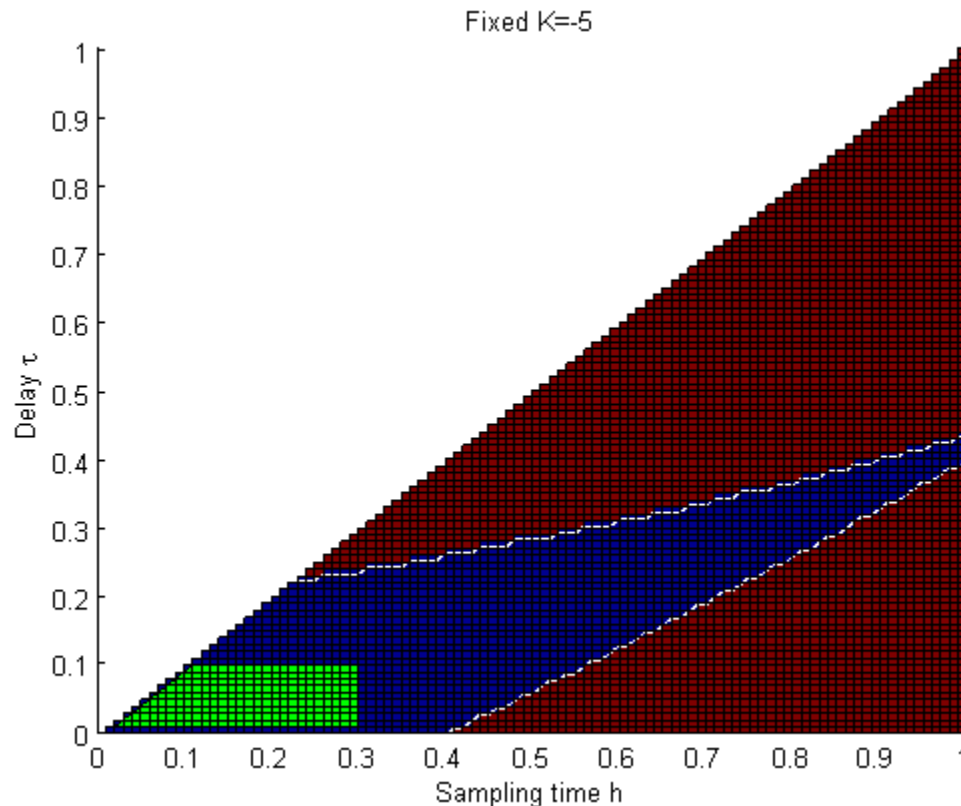
Example 3

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$$\begin{cases} \dot{x}_s(t) &= -x_s(t) + u(t) \\ y(t) &= x_s(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\Phi(h_k, \tau_k) = \begin{bmatrix} e^{-h_k} + K(1 - e^{-(h_k - \tau_k)}) & Ke^{-(h_k - \tau_k)}(1 - e^{-\tau_k}) \\ 1 & 0 \end{bmatrix}$$



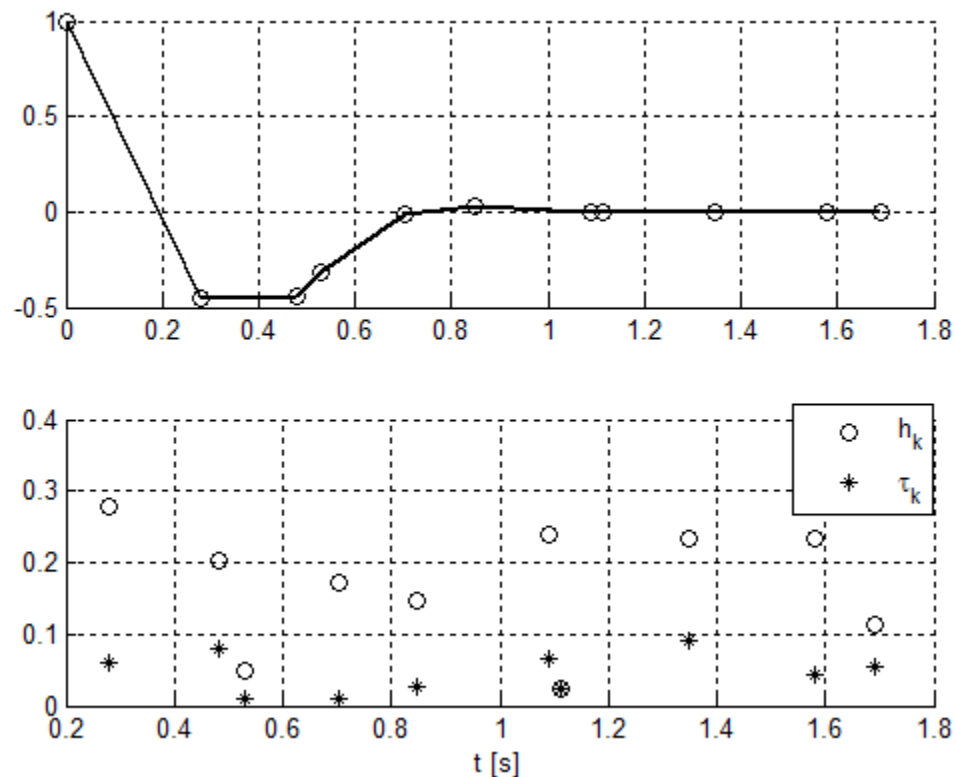
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$$\Phi(h_k, \tau_k) = \begin{bmatrix} e^{-h_k} + K(1 - e^{-(h_k - \tau_k)}) & Ke^{-(h_k - \tau_k)}(1 - e^{-\tau_k}) \\ 1 & 0 \end{bmatrix}$$



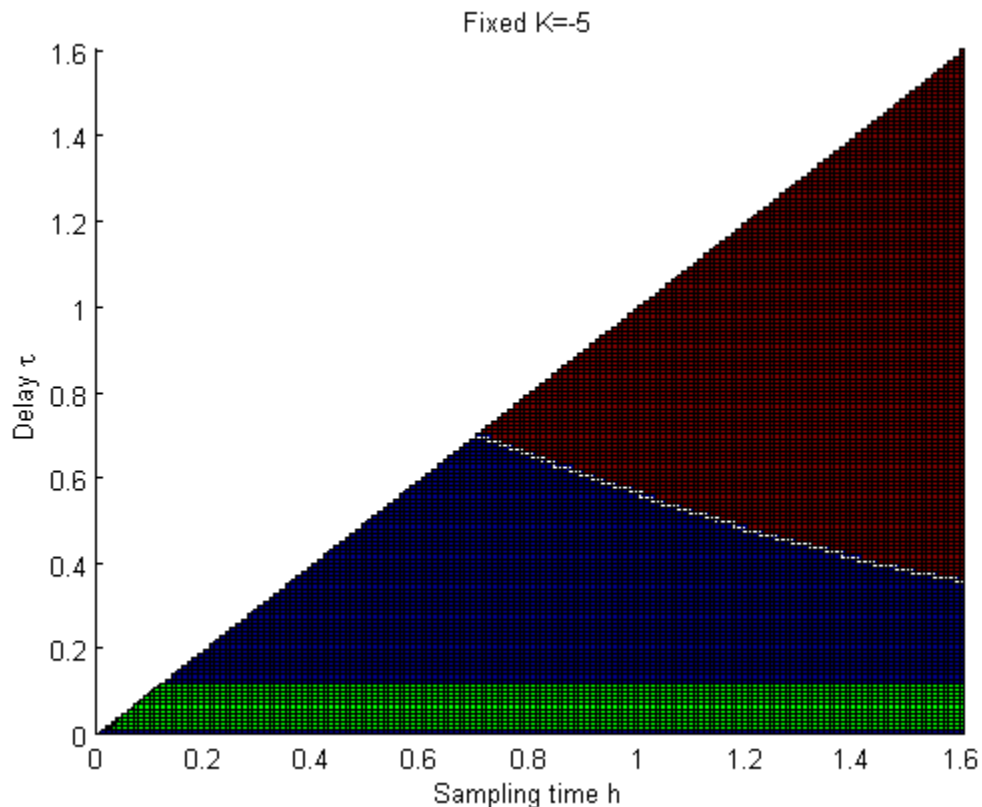
Example 4

The dynamics of the 2x2 system $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) = A_S x_S(t) + B_S u(t) \\ y(t) = C_S x_S(t) \end{cases}, A_S = \begin{bmatrix} -1 & 0.9 \\ 0.2 & -0.2 \end{bmatrix}, B_S = \begin{bmatrix} 1 \\ 0.1 \end{bmatrix}, C_S = [-0.1 \ 1]$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\Phi(h_k, \tau_k) = \begin{bmatrix} e^{A_S h_k} + A_S^{-1}(e^{A_S(h_k - \tau_k)} - I)B_S K C_S & e^{A_S(h_k - \tau_k)} A_S^{-1}(e^{A_S \tau_k} - I)B_S K \\ C_S & 0 \end{bmatrix}$$



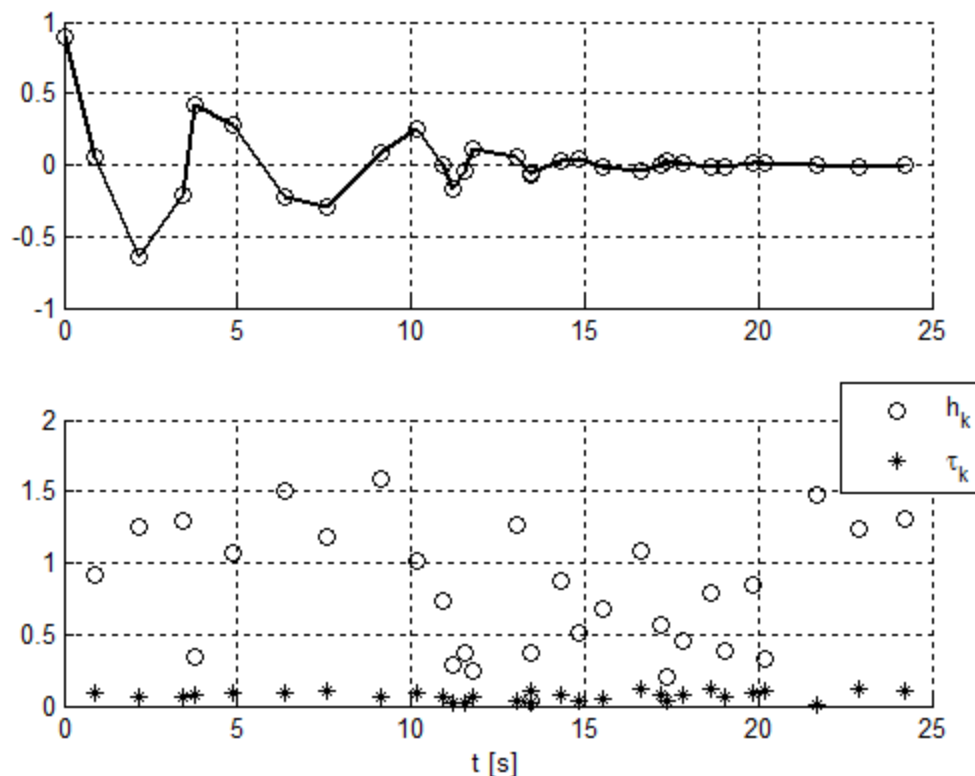
Example 4

The dynamics of the 2x2 system $\mathcal{S}$ is

$$\begin{cases} \dot{x}_S(t) = A_S x_S(t) + B_S u(t) \\ y(t) = C_S x_S(t) \end{cases}, A_S = \begin{bmatrix} -1 & 0.9 \\ 0.2 & -0.2 \end{bmatrix}, B_S = \begin{bmatrix} 1 \\ 0.1 \end{bmatrix}, C_S = \begin{bmatrix} -0.1 & 1 \end{bmatrix}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

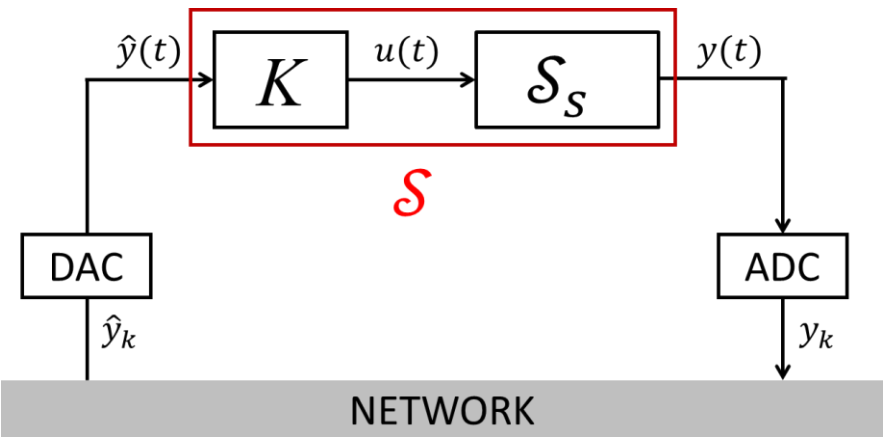
$$\Phi(h_k, \tau_k) = \begin{bmatrix} e^{A_S h_k} + A_S^{-1}(e^{A_S(h_k - \tau_k)} - I)B_S K C_S & e^{A_S(h_k - \tau_k)} A_S^{-1}(e^{A_S \tau_k} - I)B_S K \\ C_S & 0 \end{bmatrix}$$



Outline

- Reference configuration
- Examples (constant delay)
- Examples (time-varying sampling time and delay)
- **Compensation of delays**
- Examples (constant sampling time and delay, with packet dropouts)

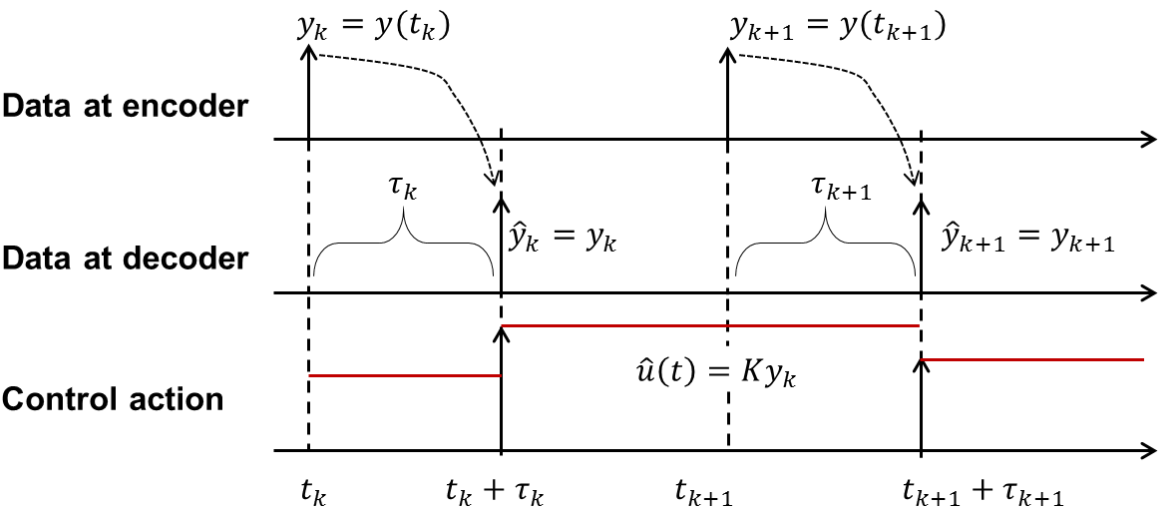
Compensation of delays (controller co-located with actuator)



The dynamics of $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) &= A_S x_S(t) + B_S u(t) \\ y(t) &= C_S x_S(t) \end{cases}$$

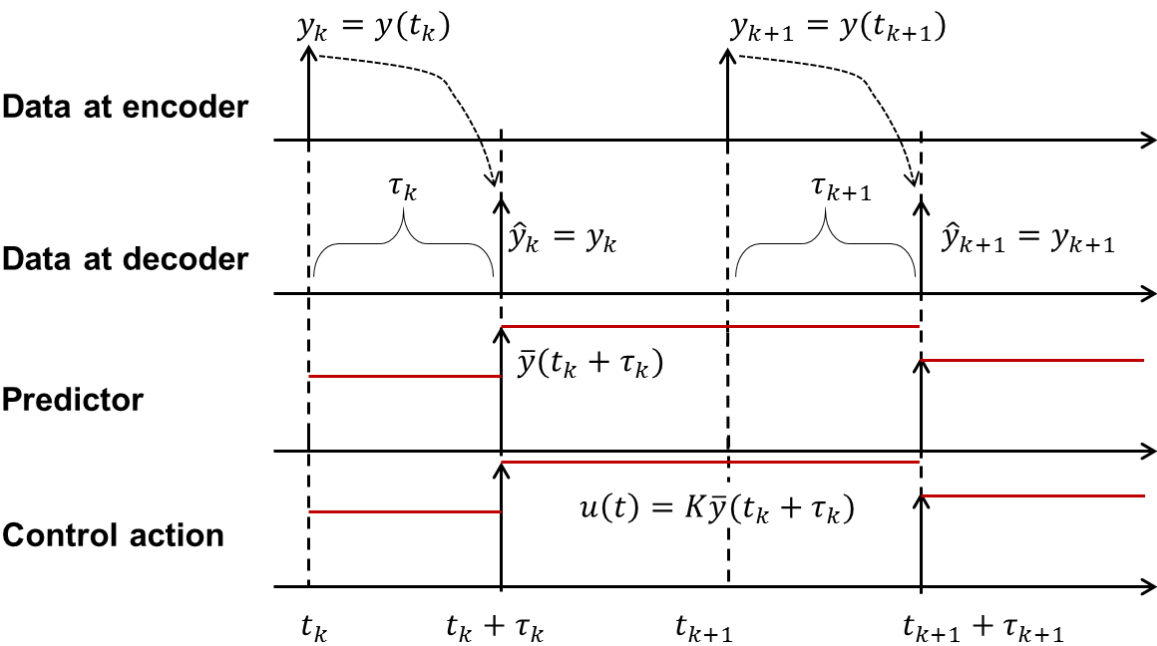
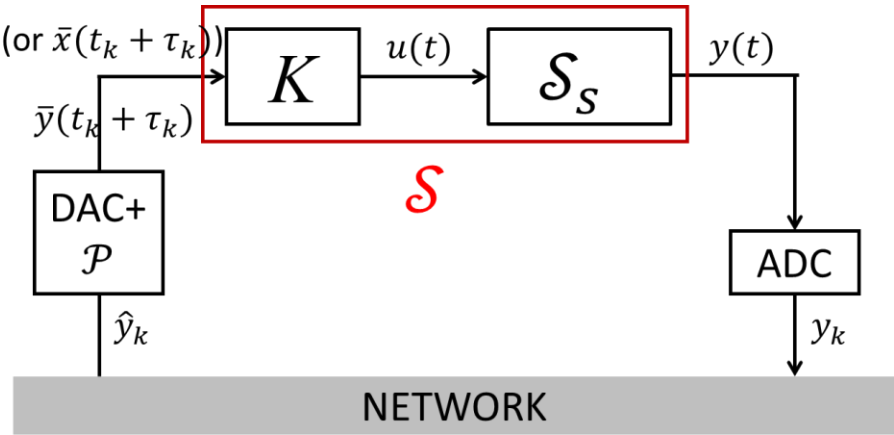
In case time stamps are not used (**unknown delay**)



- $u(t) = Ky(t_k)$ is applied in the interval $[t_k + \tau_k, t_{k+1} + \tau_{k+1})$
- Constant for $h_k^* = h_k + \tau_{k+1} - \tau_k$ s

Compensation of delays (controller co-located with actuator)

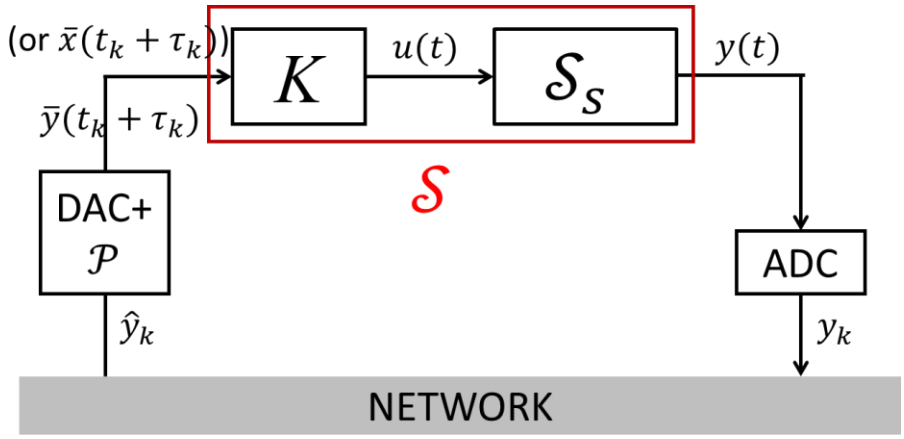
In case time stamps are used (**known delay**) a predictor $\mathcal{P}$ can be introduced:



- $u(t) = K\bar{y}(t_k + \tau_k)$ is applied in the interval $[t_k + \tau_k, t_{k+1} + \tau_{k+1})$
- Constant for $h_k^* = h_k + \tau_{k+1} - \tau_k$ s

Compensation of delays (controller co-located with actuator)

In case time stamps are used (**known delay**) a predictor $\mathcal{P}$ can be introduced:



Standard filter. Digital observer of the system state $x_k = x(t_k)$:

$$\bar{x}_{k|k} = e^{A_S h_{k-1}} \bar{x}_{k-1} + e^{A_S (h_{k-1} - \tau_{k-1})} \Gamma(\tau_{k-1}) B_S \hat{u}_{k-2} + \Gamma(h_{k-1} - \tau_{k-1}) B_S \hat{u}_{k-1} \quad \text{prediction}$$

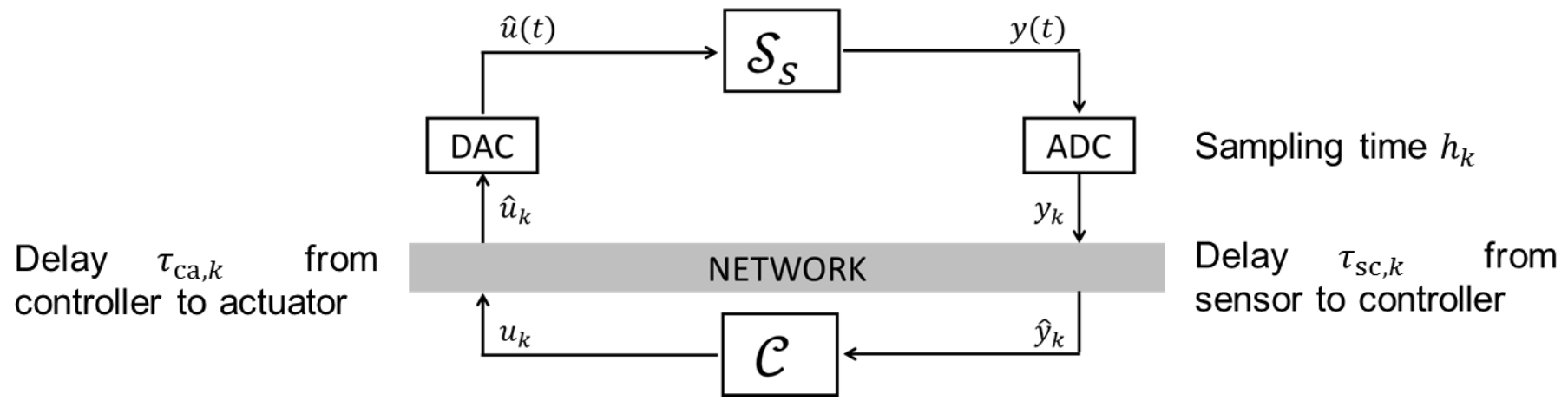
$$\bar{x}_k = \bar{x}_{k|k} + L(y_k - C_S \bar{x}_{k|k}) \quad \text{correction}$$

+ **prediction** (with step τ_k):

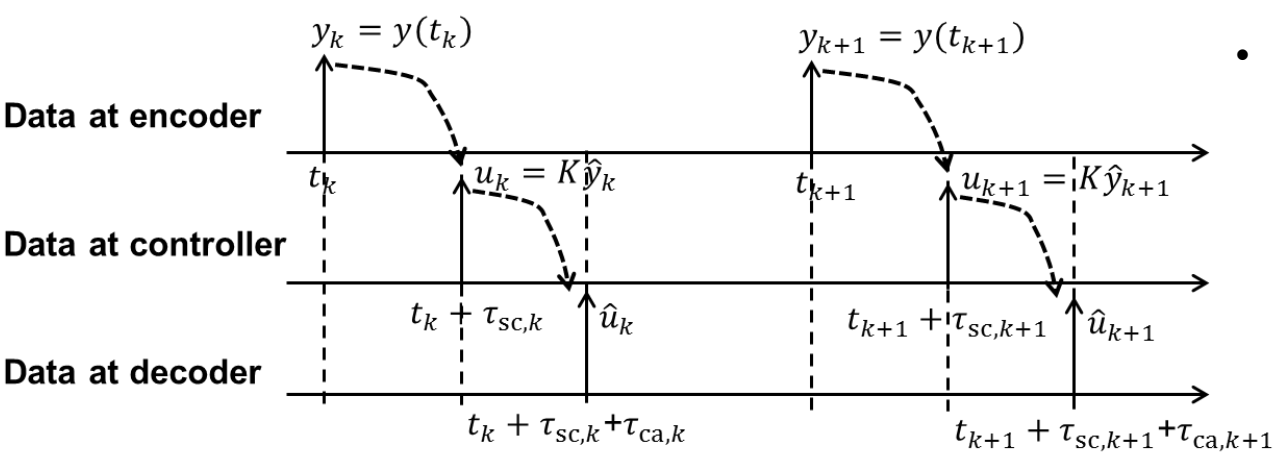
$$\begin{aligned} \bar{x}(t_k + \tau_k) &= e^{A_S \tau_k} \bar{x}_k + \Gamma(\tau_k) B_S \hat{u}_{k-1} \\ \bar{y}(t_k + \tau_k) &= C_S \bar{x}(t_k + \tau_k) \end{aligned}$$

Remark: also state-feedback control laws can be implemented using $\bar{x}(t_k + \tau_k)$.

Compensation of delays (remote controller)



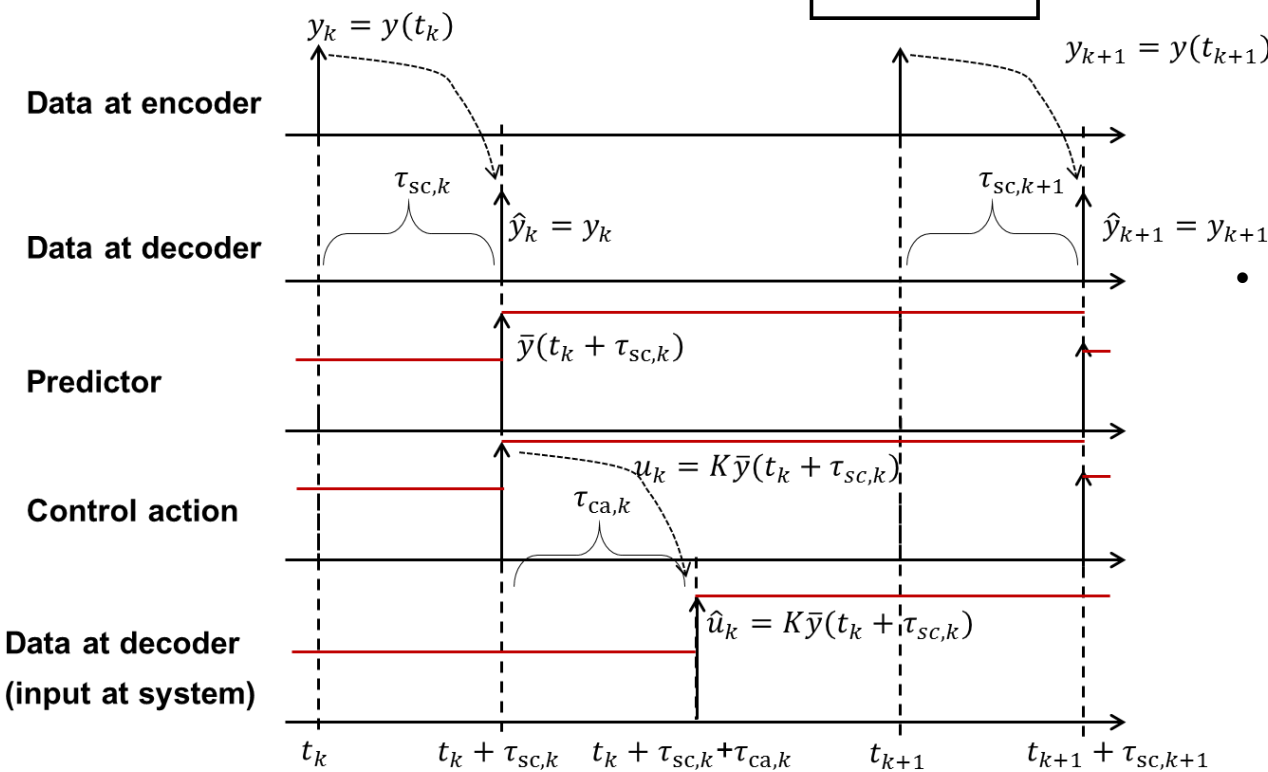
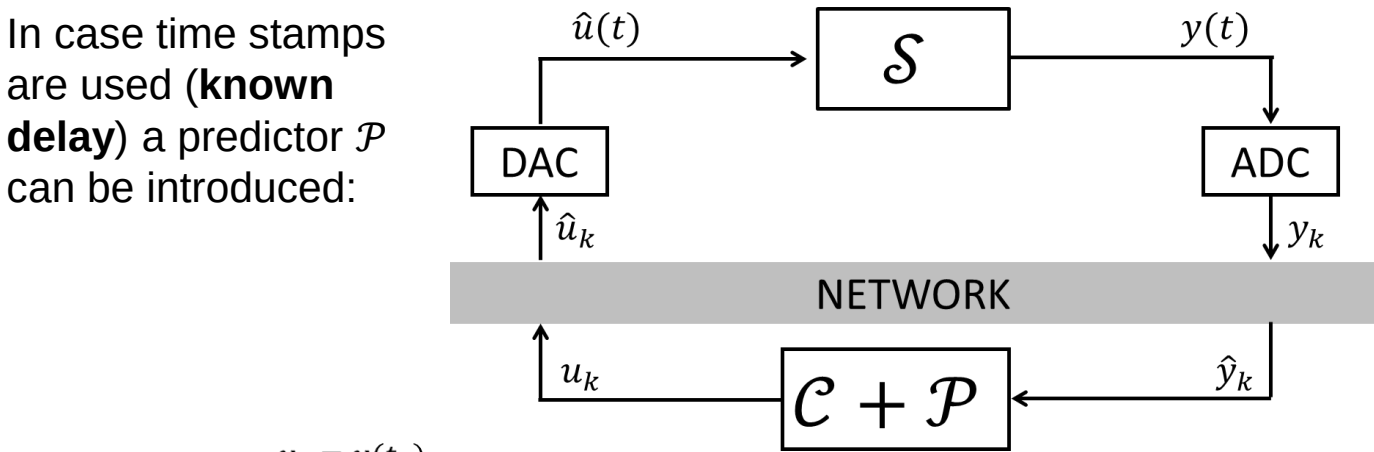
In case time stamps are not used (**unknown delay**)



- $u(t) = Ky(t_k)$ is applied in the interval $[t_k + \tau_{sc,k} +$

Compensation of delays (remote controller)

In case time stamps are used (**known delay**) a predictor $\mathcal{P}$ can be introduced:



- $u(t) = K\bar{y}(t_k + \tau_{sc,k})$ is applied in the interval $[t_k + \tau_{sc,k} + \tau_{ca,k}, t_{k+1} + \tau_{sc,k+1}]$

Outline

- Reference configuration
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- Compensation of delays
- **Examples (constant sampling time and delay, with packet dropouts)**

Example 1

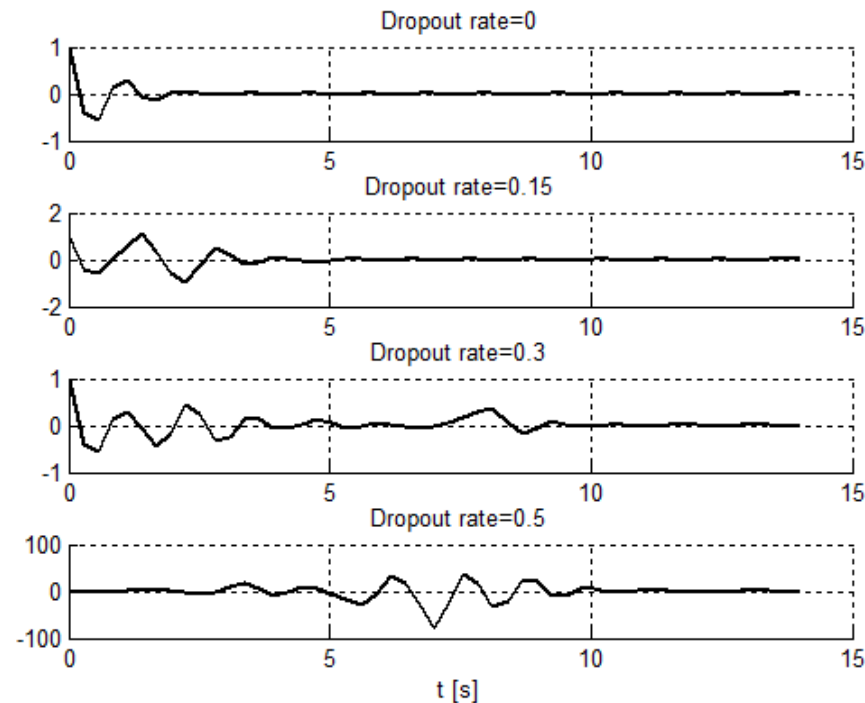
The dynamics of the scalar system $\mathcal{S}_s$ is

$$\begin{cases} \dot{x}_s(t) &= u(t) \\ y(t) &= x_s(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\tilde{\Phi}_1 = \Phi(h, \tau) = \begin{bmatrix} 1 + K(h - \tau) & K\tau \\ 1 & 0 \end{bmatrix}$$
$$\tilde{\Phi}_0 = \begin{bmatrix} 1 & Kh \\ 0 & I \end{bmatrix}$$

$$\bar{r} = 0.2802, \bar{p} = 0.23$$



Example 2

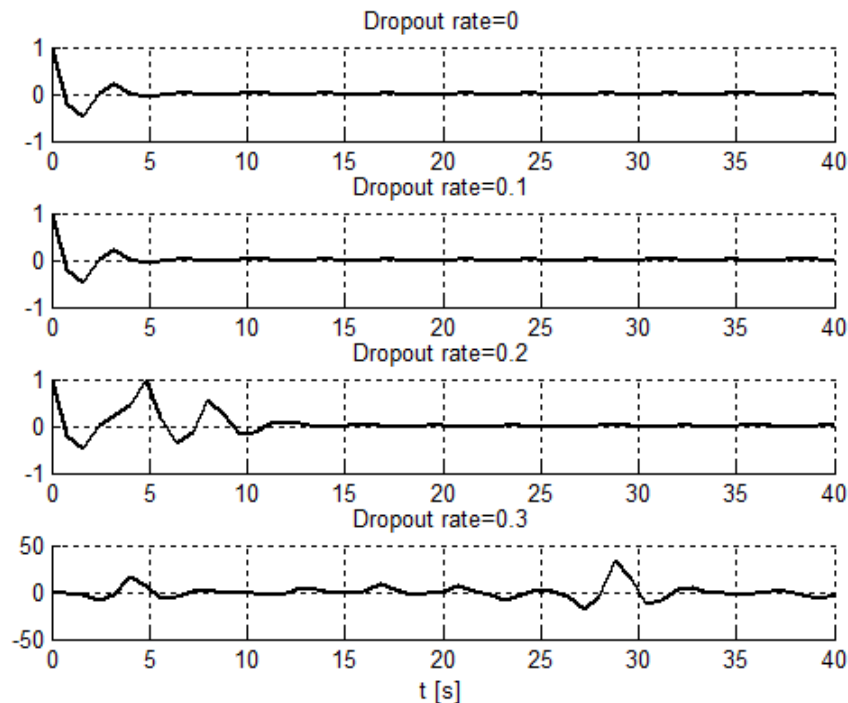
The dynamics of the scalar system $\mathcal{S}_s$ is

$$\begin{cases} \dot{x}_s(t) &= x_s(t) + u(t) \\ y(t) &= x_s(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\tilde{\Phi}_1 = \begin{bmatrix} e^h + K(e^{(h-\tau)} - 1) & Ke^{(h-\tau)}(e^\tau - 1) \\ 1 & 0 \end{bmatrix}, \tilde{\Phi}_0 = \begin{bmatrix} e^h & K(e^h - 1) \\ 0 & 1 \end{bmatrix}$$

$$\bar{r} = 0.2224, \bar{p} = 0.08$$



Example 3

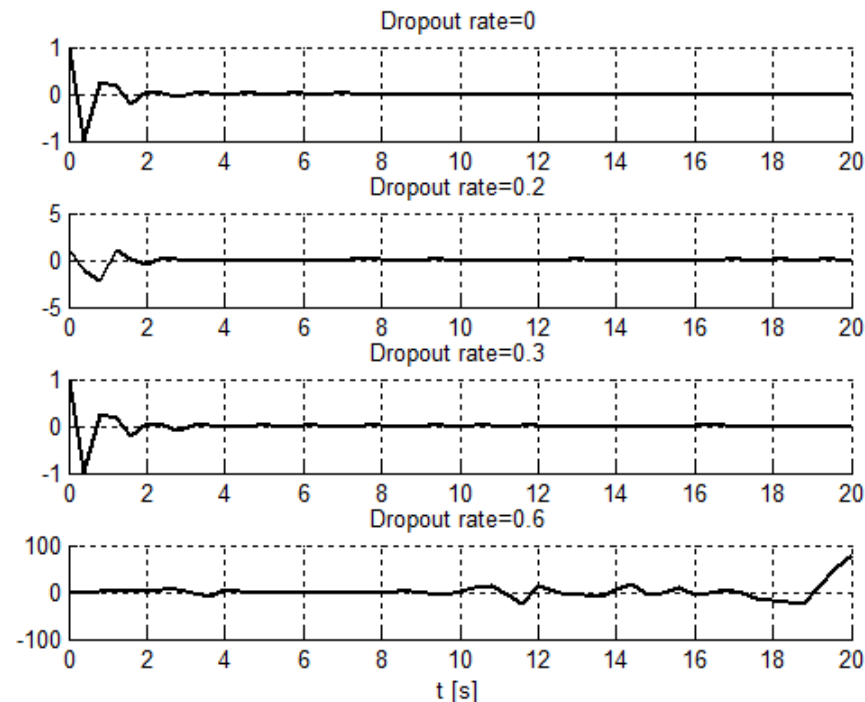
The dynamics of the scalar system $\mathcal{S}_s$ is

$$\begin{cases} \dot{x}_s(t) &= -x_s(t) + u(t) \\ y(t) &= x_s(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\tilde{\Phi}_1 = \begin{bmatrix} e^{-h} + K(1 - e^{-(h-\tau)}) & Ke^{-(h-\tau)}(1 - e^{-\tau}) \\ 1 & 0 \end{bmatrix}, \tilde{\Phi}_0 = \begin{bmatrix} e^{-h} & K(1 - e^{-h}) \\ 0 & 1 \end{bmatrix}$$

$$\bar{r} = 0.2975, p = 0.19$$



Example 4

The dynamics of the 2x2 system $\mathcal{S}$ is

$$\begin{cases} \dot{x}_S(t) = A_S x_S(t) + B_S u(t) \\ y(t) = C_S x_S(t) \end{cases}, A_S = \begin{bmatrix} -1 & 0.9 \\ 0.2 & -0.2 \end{bmatrix}, B_S = \begin{bmatrix} 1 \\ 0.1 \end{bmatrix}, C_S = [-0.1 \ 1]$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\tilde{\Phi}_1 = \begin{bmatrix} e^{A_S h} + A_S^{-1}(e^{A_S(h-\tau)} - I)B_S K C_S & e^{A_S(h-\tau)} A_S^{-1}(e^{A_S \tau} - I)B_S K \\ C_S & 0 \end{bmatrix}$$

$$\tilde{\Phi}_0 = \begin{bmatrix} e^{A_S h} & A_S^{-1}(e^{A_S h} - I)B_S K \\ 0 & 1 \end{bmatrix}$$

$$\bar{r} = 0.1119, \bar{p} = 0.11$$

